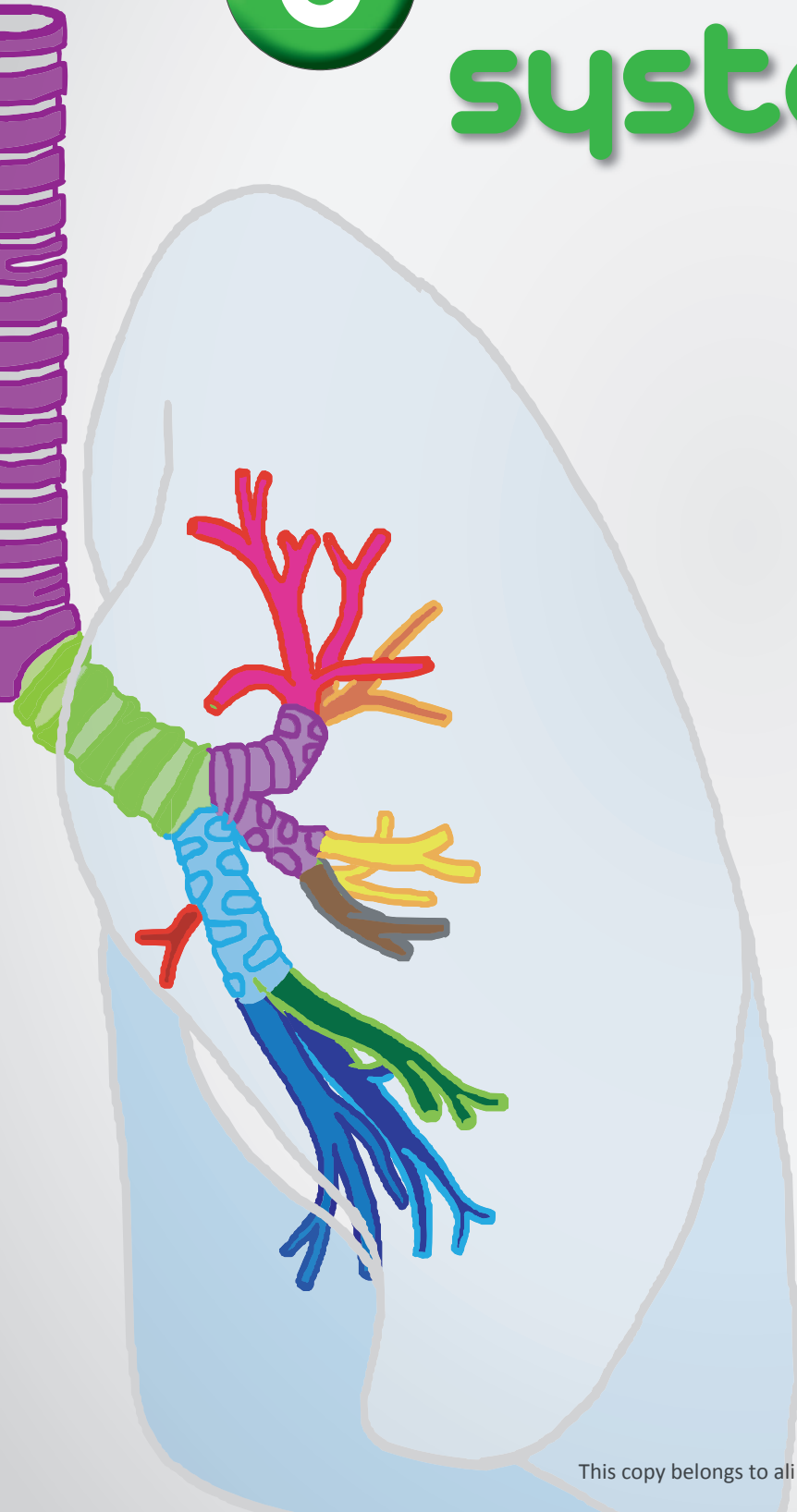


Memorix Anatomy

6

Respiratory system

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The **respiratory tract** is a **tree of ducts**, which is divided into the **upper and lower airways**. Anatomically, the upper respiratory tract ends with the laryngopharynx; clinically, it includes the larynx and the cervical part of the trachea. The wall of the airways is strengthened by cartilage and elastic fibres, which protect the airway from collapse and provide elasticity. The respiratory system is responsible for the **exchange of gases** between the **air and blood**, **voice production**, **maintenance of acid-base balance** and **elimination of substances**.

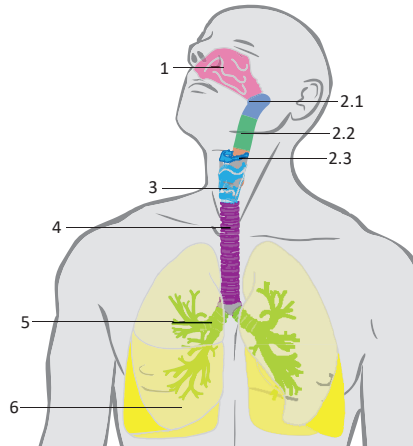
Basic parts

Upper respiratory tract

- 1 **Nasal cavity** (*cavitas nasi*)
 - 1.1 **Paranasal sinuses** (*sinus paranasales*)
- 2 **Pharynx**
 - 2.1 **Nasopharynx** (*pars nasalis pharyngis*)
 - 2.2 **Oropharynx** (*pars oralis pharyngis*)
 - 2.3 **Laryngopharynx or hypopharynx** (*pars laryngea pharyngis*)

Lower respiratory tract

- 3 **Larynx**
- 4 **Trachea**
- 5 **Bronchi**
- 6 **Lungs** (*pulmones*)



Rhinos is the Greek term for nose.

During **ontogenetic development**, the ventral aspect of the foregut forms the laryngotracheal groove, which gives rise to the larynx, trachea, bronchi and lungs.

The **airways** are almost entirely lined by mucosa with pseudostratified ciliated columnar epithelium. Coordinated ciliary movements ("mucociliary escalator") beat a layer of secreted mucus towards the pharynx, cleaning the airways.

The **olfactory region** (*regio olfactoria*) comprises the mucosa of the ceiling of the nasal cavity and the upper third of the nasal septum. It is lined with olfactory epithelium, which is specialised sensory pseudostratified ciliated columnar epithelium. The supporting cells contain lipofuscin granules, imparting a yellowish colour to the mucosa.

The **respiratory region** (*regio respiratoria*) constitutes all the mucosa of the nasal cavity excluding the olfactory region. It is lined by pseudostratified ciliated columnar epithelium.

The **nasal vestibule** contains thick hairs (*vibrissae*), which mechanically prevent particles from entering the nasal cavity.

The **vomer nasal cartilage** (*cartilago vomeronasalis*) is a minor paired cartilage, which is fixed to the septal nasal cartilage. It is rudimentary and insignificant in humans.

In **plastic surgery**, the **columella** is described as the skin that overlies the outer margin of the nasal septum.

Clinical notes

The **nasal cavity can be surgically approached** externally (the classic approach) or endoscopically and microscopically (functional endoscopic sinus surgery).

The **pituitary gland** may be surgically approached through the nasal cavity and sphenoidal sinus.

A **deviated nasal septum** may, during inflammation or oedema, obstruct half of the nasal cavity or block the openings of the paranasal sinuses.

Liquorrhoea is leakage of cerebrospinal fluid from the subarachnoid space. In fractures or surgeries affecting the anterior cranial fossa, cerebrospinal fluid may leak from the nose (nasal liquorrhoea).

The nose has a **bony and cartilaginous base** of a triangular shape. The skin of the nose is **rich in sebaceous glands**. The skin overlying the bony part is movable, while the skin overlying cartilaginous part is firmly fixed. The muscles of facial expression inserting into the nose constrict and flare the nostrils.

Basic parts

- 1 **Root of nose** (*radix nasi*)
- 2 **Dorsum of nose / bridge of nose** (*dorsum nasi*)
- 3 **Apex of nose / tip of nose** (*apex nasi*)
- 4 **Ala of nose / wing of nose** (*ala nasi*)
- 5 **Nostrils** (*nares*)
- 6 **Nasal septum** (*septum nasi*)

Structure

- 1 **Nasal bone** (*os nasale dextrum et sinistrum*)
- 2 **Nasal cartilages** (*cartilagine nasii*)
 - 2.1 **Lateral nasal cartilage** (*cartilago nasi lateralis*)
 - 2.2 **Major alar cartilage with medial and lateral crus** (*cartilago alaris major – crus mediale et laterale*)
 - 2.3 **Minor alar cartilages** (*cartilagine alares minores*)
 - 2.4 **Septal nasal cartilage** (*cartilago septi nasi*)
 - 2.5 **Accessory nasal cartilages** (*cartilagine nasales accessoriae*)

Vascular supply and innervation

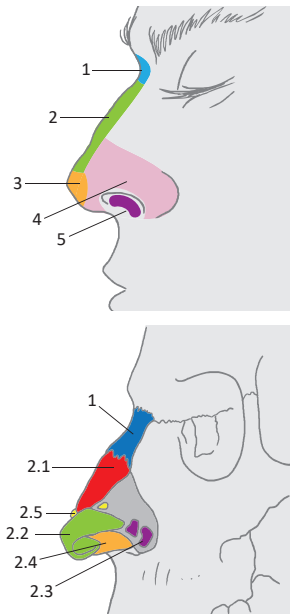
Arterial supply: angular artery (facial artery), infraorbital artery (maxillary artery), supratrochlear and infratrochlear arteries (ophthalmic artery)

Venous drainage: facial vein, inferior ophthalmic vein, pterygoid plexus

Lymphatic drainage: submandibular nodes

Somatomotor (branchiomotor): nasal muscles of facial expression (facial nerve)

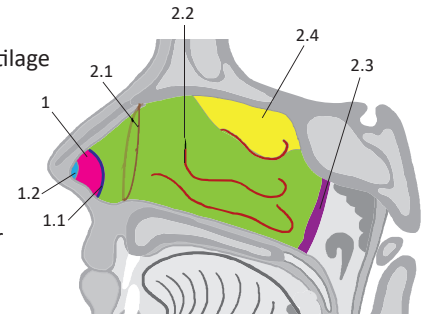
Somatosensory: ophthalmic nerve, maxillary nerve (trigeminal nerve)



The nasal cavity is open through the nostrils to the external environment. Dorsally, it communicates with the **nasopharynx** through the posterior nasal **apertures**. It is divided sagittally by the **nasal septum**, which is composed of cartilage ventrally and bone dorsally. The floor of the nasal cavity is formed by the palate. The three **conchae** (inferior, medial and superior) protrude from the lateral wall and divide the cavity into four groove-like air passages, the inferior, middle, superior and common meatuses, which communicate with the **paranasal sinuses** and the **nasolacrimal duct**. The roof is formed by the base of the anterior cranial fossa and is involved in **olfaction**. The nasal cavity serves to **heat and humidify inspired air**, and mechanically blocks particles from entering.

Basic parts

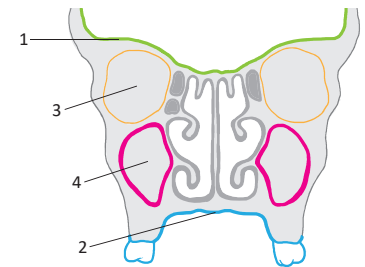
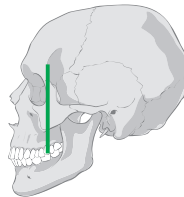
- 1 **Nasal vestibule** (*vestibulum nasi*) – from the nostrils to the upper edge of the major alar cartilage
 - 1.1 **Limen nasi** – the border between the nasal vestibule and the nasal cavity proper
 - a ridge formed by the upper edge of the major alar cartilage
 - 1.2 **Apical recess** (*recessus apicis nasi*)
 - a ventral projection (cul-de-sac) of the vestibule into the nasal apex
- 2 **Nasal cavity proper** (*cavitas nasi propria*)
 - 2.1 **Piriform aperture** (*apertura piriformis*) – the bony entrance of the nasal cavity proper
 - 2.2 **Nasal conchae/turbinates** (*conchae nasales*)
 - 2.3 **Posterior nasal apertures / choanae** – communicate with the nasopharynx
 - 2.4 **Olfactory region** (*regio olfactoria*) – olfactory epithelium with the olfactory receptors



Sagittal section of the nasal cavity

Syntopy

- 1 **Cranially**: anterior cranial fossa (via cribriform plate of ethmoidal bone)
- 2 **Caudally**: oral cavity (via hard palate)
- 3 **Laterocranially**: orbit (via ethmoidal bone)
- 4 **Laterocaudally**: maxillary sinus
- 5 **Ventrally**: nose
- 6 **Dorsally**: nasopharynx (via choanae)



Frontal section of the nasal cavity

Borders

1 Roof

- 1.1 **Lateral nasal cartilage** (*cartilago nasi lateralis*)
- 1.2 **Nasal bone** (*os nasale*)
- 1.3 **Nasal part of frontal bone** (*pars nasalis ossis frontalis*)
- 1.4 **Cribriform plate of ethmoidal bone** (*lamina cribrosa ossis ethmoidalis*)
- 1.5 **Body of sphenoid** (*corpus ossis sphenoidalis*)

2 Floor

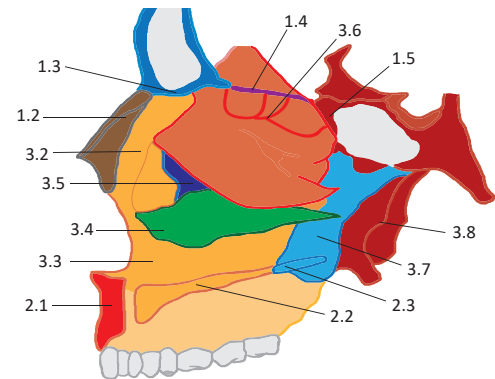
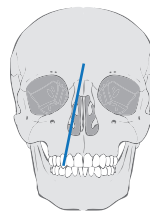
- 2.1 **Incisive bone** (*premaxilla / os incisivum*)
- 2.2 **Palatine process of maxilla** (*processus palatinus maxillae*)
- 2.3 **Horizontal plate of palatine bone** (*lamina horizontalis ossis palatini*)

3 Lateral wall

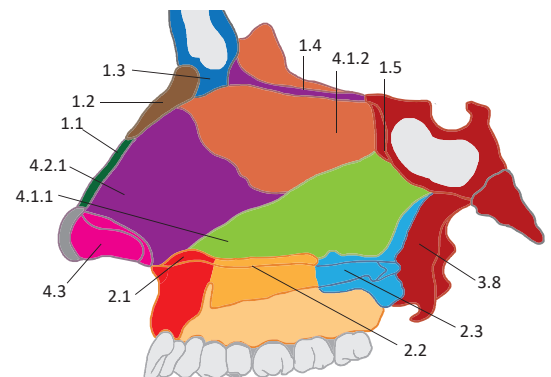
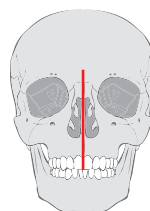
- 3.1 **Cartilaginous nasal wall**
- 3.2 **Frontal process of maxilla** (*processus frontalis maxillae*)
- 3.3 **Body of maxilla** (*corpus maxillae*)
- 3.4 **Inferior nasal concha** (*concha nasalis inferior*) – fixed to the maxilla
- 3.5 **Lacrimal bone** (*os lacrimale*)
- 3.6 **Ethmoidal labyrinth** (*labyrinthus ethmoidalis*)
- 3.7 **Perpendicular plate of palatine bone** (*lamina perpendicularis ossis palatini*)
- 3.8 **Medial plate of pterygoid process of sphenoid** (*lamina medialis processus pterygoidei ossis sphenoidalis*)

4 Medial wall – nasal septum (*septum nasi*)

- 4.1 **Bony part** (*pars ossea*)
 - 4.1.1 **Vomer**
 - 4.1.2 **Perpendicular plate of ethmoid** (*lamina perpendicularis ossis ethmoidalis*)
- 4.2 **Cartilaginous part** (*pars cartilaginea*)
 - 4.2.1 **Septal nasal cartilage** (*cartilago septi nasi*)
- 4.3 **Membranous part** (*pars membranacea*) – fibrous part between the nostrils
- 4.4 **Cutaneous part** (*pars cutanea*) – outer part between the nostrils



Oblique section of the nasal cavity



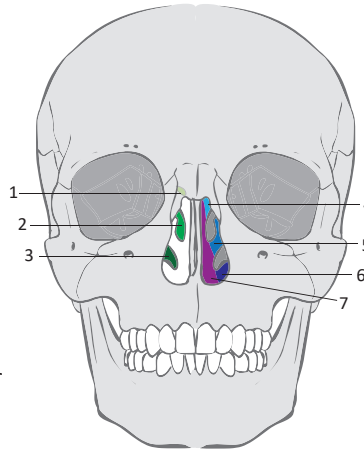
Sagittal view of the nasal septum

Nasal conchae or turbinates (*conchae nasales*)

- 1 **Superior nasal concha** (*concha nasalis superior*) – the smallest concha, – a part of the ethmoidal bone
- 2 **Middle nasal concha** (*concha nasalis media*) – a part of the ethmoidal bone
- 3 **Inferior nasal concha** (*concha nasalis inferior*) – a separate bone joined to the maxilla

Nasal meatus (*meatus nasi*)

- 4 **Superior nasal meatus** (*meatus nasi superior*)
 - is located between the roof of the nasal cavity and the middle nasal concha
 - contains the openings of the sphenoid sinus and posterior ethmoidal air cells
- 5 **Middle nasal meatus** (*meatus nasi medius*)
 - is located between the middle and inferior nasal conchae
 - contains the openings of the frontal sinus, maxillary sinus and anterior and middle ethmoidal air cells
- 6 **Inferior nasal meatus** (*meatus nasi inferior*)
 - is located between the inferior nasal concha and the floor of the nasal cavity
 - contains the opening of the nasolacrimal duct
- 7 **Common nasal meatus** (*meatus nasi communis*) – is located between the medial ends of the conchae and the nasal septum
- 8 **Nasopharyngeal meatus** (*meatus nasopharyngeus*)
 - a space between the dorsal ends of the conchae and the choanae



Vascular supply

Arterial supply: anterior and posterior ethmoidal arteries (ophthalmic artery), descending palatine artery (maxillary artery), sphenopalatine artery (maxillary artery), facial artery

Venous drainage: facial vein, ethmoidal veins, pterygoid plexus, pharyngeal plexus

Lymphatic drainage: retropharyngeal nodes and deep anterior cervical nodes

Innervation

Somatosensory: anterior ethmoidal nerve (ophthalmic nerve), posterior nasal branches and infraorbital nerve (maxillary nerve)

Visceromotor sympathetic: periarterial plexuses and deep petrosal nerve

Visceromotor parasympathetic: facial nerve

Special sensory: olfactory nerve

The **semilunar hiatus** is the opening of the maxillary sinus, frontal sinus and anterior and middle ethmoidal air cells into the middle nasal meatus.

The **sphenoethmoidal recess** is the dorsocranial projection of the nasal cavity towards the upper end of the anterior aspect of the sphenoid bone. The sphenoidal sinus opens here through the superior nasal meatus.

The **cavernous plexuses of the conchae** are vascular plexuses on the conchae.

Kiesselbach's plexus is located on the cartilaginous septum near the anterior end of the inferior nasal concha. It is a frequent source of nasal bleeding (epistaxis). It is supplied by 5 arteries: superior labial, anterior and posterior ethmoidal, sphenopalatine and greater palatine.

Woodruff's plexus is a venous plexus on the dorsal end of the inferior nasal concha, which may be a source of venous bleeding.

The **paranasal sinuses** arise as outgrowths of the nasal mucosa into the neighbouring bones. They are therefore covered by mucosa with respiratory epithelium. They develop mostly after birth; the newborn has only very rudimentary ethmoidal air cells and maxillary sinuses.

Clinical notes

During inflammation of the paranasal sinuses (sinusitis), percussion over the sinus becomes painful. The inflammation may erode the thin bone and enter the cranial cavity or the orbit. An X-ray may show opacity of the sinus.

During inflammation of the frontal sinus (frontal sinusitis), palpation of the ophthalmic nerve (N. V₁) may produce pain

During inflammation of the maxillary sinus (maxillary sinusitis), palpation of the maxillary nerve (N. V₂) may produce pain.

During inflammation of the ethmoidal air cells, the nasal root may be tender to palpation.

The **maxillary sinus** is closely related to the roots of the upper teeth, especially the canine, which has the longest root.

The **ethmoidal air cells** are separated from the orbit by the very thin orbital plate of the ethmoid (*lamina orbitalis ossis ethmoidalis* or, the obsolete term, *lamina papyracea*). Infection may, especially in young children, erode the thin bone and spread into the orbit and cause orbital cellulitis.

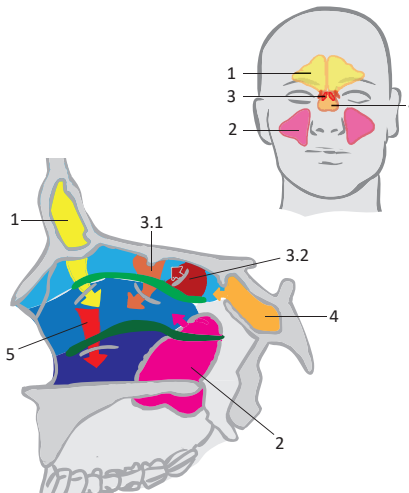
The paranasal sinuses are air filled cavities within the bones of the skull. They **increase the resonance of voice** and **decrease the weight of the skull**.

Paranasal sinuses

- 1 **Frontal sinus** (*sinus frontalis*)
 - opens into the middle nasal meatus
- 2 **Maxillary sinus** (*sinus maxillaris*)
 - opens into the middle nasal meatus
- 3 **Ethmoidal air cells** (*cellulae ethmoidales*)
 - 3.1 **Anterior and middle ethmoidal air cells** (*cellulae ethmoidales anteriores et mediae*)
 - open into the middle nasal meatus
 - 3.2 **Posterior ethmoidal air cells** (*cellulae ethmoidales posteriores*)
 - open into the superior nasal meatus
- 4 **Sphenoid sinus** (*sinus sphenoidalis*)
 - opens into the superior nasal meatus

Lacrimal drainage

- 5 **Nasolacrimal duct** (*ductus nasolacrimalis*)
 - drains into the inferior nasal meatus



The pharynx is a muscular funnel-shaped organ attached to the base of the skull. Caudally it **continues as the oesophagus**. Ventrally it communicates with the nasal, oral and laryngeal cavities. The oral and laryngeal parts of the pharynx are components of both the gastrointestinal and respiratory tracts. The function of the pharynx is based on **two types of paired muscles: levators and constrictors**. The pharynx has an important role in **swallowing, speaking and breathing**.

Division and communications

- 1 **Nasopharynx** (*pars nasalis pharyngis*)
 - communicates with the nasal cavity through the choanae
- 2 **Oropharynx** (*pars oralis pharyngis*)
 - communicates with the oral cavity through the isthmus of fauces
- 3 **Laryngopharynx** (*pars laryngea pharyngis*)
 - communicates with the laryngeal cavity through the laryngeal inlet

Syntopy

- 4 **Cranially**: cranial base
- 5 **Caudally**: at the level of the vertebra C6 continues as the oesophagus
- 6 **Ventrally**: opens into the nasal, oral and laryngeal cavities
- 7 **Dorsally**: vertebral column

Nasopharynx (*epipharynx* / *pars nasalis pharyngis*)

Borders

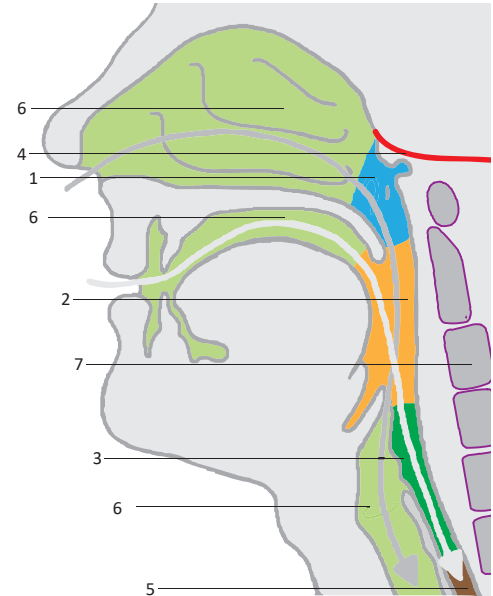
Cranially: attached to the cranial base

Caudally: extends to the dorsal part of the soft palate

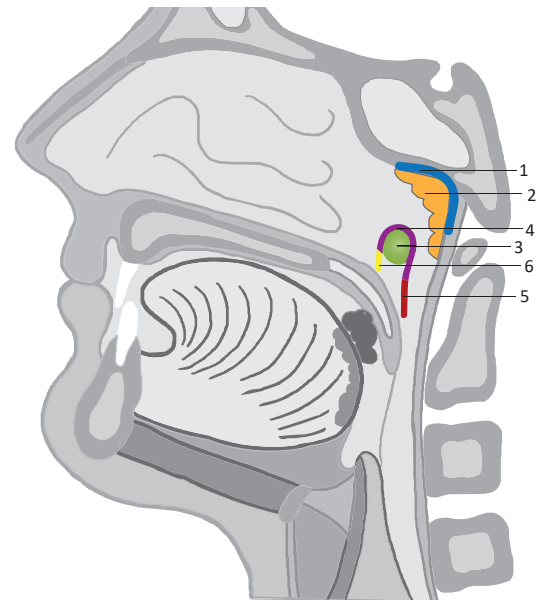
Ventrally: communicates with the nasal cavity through the choanae

Structures

- 1 **Vault of pharynx** (*fornix pharyngis*)
 - is attached to the cranial base
 - has a ventrally opened concavity
- 2 **Pharyngeal tonsil** (*tonsilla pharyngea*)
 - an unpaired tonsil in the mucosa of the dorsocranial part of the vault of the pharynx
- 3 **Pharyngeal opening of auditory tuba** (*ostium pharyngeum tubae auditivae*)
 - the opening of the auditory tube on the lateral side of the nasopharynx
- 4 **Torus tubarius** – an elevation of mucosa behind the opening of the auditory tube formed by the medial lamina of the cartilaginous part of the auditory tube
- 5 **Salpingopharyngeal fold** (*plica salpingopharyngea*)
 - a fold formed by the salpingopharyngeus muscle, extending caudally from the torus tubarius
- 6 **Salpingopalatine fold** (*plica salpingopalatina*)
 - a fold extending from the pharyngeal opening of the auditory tube to the soft palate
- 7 **Torus levatorius** – a ridge between the salpingopalatine and salpingopharyngeal folds beneath the opening of the auditory tube, formed by the levator veli palatini
- 8 **Tubal tonsil** (*tonsilla tubaria*)
 - a paired tonsil around the opening of the auditory tube



Sagittal section of the head and neck



Sagittal section of the head and neck

Oropharynx (*pars oralis pharyngis*)

Cranially: the dorsal end of the soft palate (at the level of the vertebra C2)

Caudally: the upper edge of the epiglottis (at the level of the vertebra C4)

Ventrally: opens into the oral cavity through the isthmus of fauces

Laryngopharynx (*hypopharynx* / *pars laryngea pharyngis*)

Cranially: the upper edge of the epiglottis

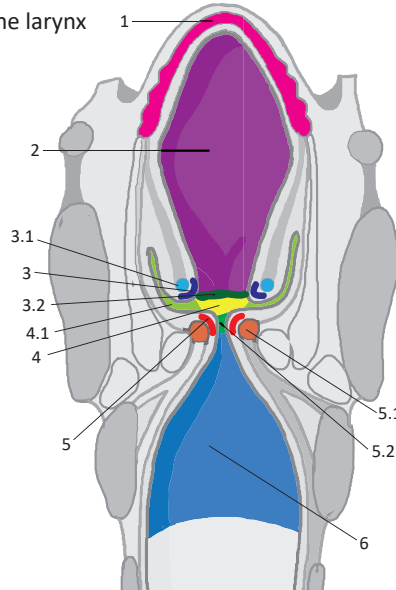
Caudally: ends at the level of the vertebra C6, where it continues as the esophagus

Ventrally: opens into the laryngeal cavity through the laryngeal inlet

The larynx is a hollow organ attached to the hyoid bone. It communicates cranially with the laryngopharynx and caudally it continues as the trachea. The laryngeal skeleton consists of a system of cartilages connected by ligaments and joints. The movement of the cartilages is provided by skeletal muscles. The larynx has two surfaces: an inner surface covered by mucosa and an outer surface covered by adventitia. The larynx plays a role in **breathing** and is the main organ of **phonation**.

Laryngeal cavity (*cavitas laryngis*)

- 1 **Laryngeal inlet** (*aditus laryngis*) – an opening in the ventral wall of the laryngopharynx that connects the pharynx to the larynx
- 2 **Laryngeal vestibule** (*vestibulum laryngis*) – a space between the laryngeal inlet and the rima vestibuli
- 3 **Vestibular folds** (*plicae vestibulares*)
 - paired mucosal folds
 - 3.1 **Vestibular ligaments / false vocal cords** (*ligamenta vestibularia*)
 - ligaments forming the vestibular folds
 - 3.2 **Rima vestibuli** (*rima vestibuli*)
 - a space between the vestibular folds
- 4 **Glottis** – a space between the vestibular folds cranially and the vocal folds caudally
 - 4.1 **Laryngeal ventricle** (*ventriculus laryngis*)
 - the lateral recess of the glottis
- 5 **Vocal folds / true vocal cords** (*plicae vocales*)
 - paired mucous folds between the thyroid and arytenoid cartilages
 - 5.1 **Vocal ligaments** (*ligamenta vocalia*)
 - form the base of the vocal folds
 - 5.2 **Rima glottidis** – a space between the vocal folds
- 6 **Infraglottic cavity** (*cavitas infraglottica*)
 - a space below the rima glottidis



Frontal section of the larynx, posterior view

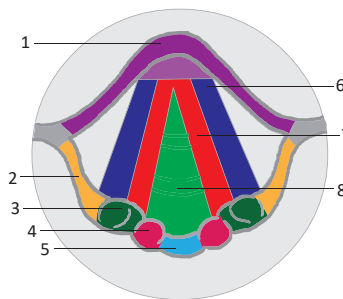
Laryngoscopic view of the glottis

Borders of the laryngeal inlet

- 1 **Epiglottis** – ventrocranially
- 2 **Aryepiglottic fold** (*plica aryepiglottica*) – laterally
- 3 **Cuneiform tubercle** (*tuberculum cuneiforme*) – laterally
- 4 **Corniculate tubercle** (*tuberculum corniculatum*) – laterally
- 5 **Interarytenoid notch** (*incisura interarytenoidea*) – dorsocaudally

Ligaments of the rima glottidis

- 6 **Vestibular folds** (*plicae vestibulares*)
- 7 **Vocal folds** (*plicae vocales*)
- 8 **Rima glottidis**



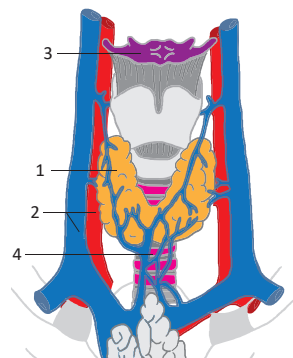
Laryngoscopic view of the glottis

Syntopy

- 1 **Ventrolaterally**: lobes of the thyroid gland
- 2 **Laterally**: common carotid arteries, internal jugular veins
- 3 **Cranially**: hyoid bone
- 4 **Caudally**: trachea
- 5 **Ventrally**: pretracheal layer of the cervical fascia
- 6 **Dorsally**: pharynx

Histological structure of the mucosa

- 1 **Respiratory epithelium** – pseudostratified ciliated columnar epithelium
 - pink mucosa in almost all parts of the larynx
- 2 **Non-keratinized stratified squamous epithelium** – whitish mucosa on the vocal folds, ary-epiglottic folds, ventral side and cranial half of the dorsal side of the epiglottis
- 3 **Zone of stratified columnar epithelium** – between the cranial and caudal halves of the dorsal side of the epiglottis



Syntopy of the larynx

The **cricoid cartilage** (*cartilago cricoidea* from the Greek word “krikoeides”, meaning “ring-shaped”) was named after its resemblance to a signet ring.

The **triticeal cartilage** (*cartilagine triticae*) is a sesamoid cartilage within the thyrohyoid ligament.

Taste receptors are located in the mucosa of the dorsal side of the epiglottis and the aryepiglottic fold.

The **anastomosis of Galen** is an anastomosis between the sensory branches of the superior laryngeal nerve and recurrent laryngeal nerve. It is located on the dorsal side of the cricoid cartilage.

The **rima glottidis** is the narrowest part of the laryngeal cavity in adults. In children, the width is almost equal to the infraglottic cavity.

The rima glottidis has three parts:

- **Intermembranous part**
 - between the vocal ligaments
- **Intercartilaginous part**
 - between the vocal processes
- **Interarytenoid part**
 - between the arytenoid cartilages

Voice change (voice mutation) in males is caused by increased production of testosterone resulting in rapid growth of the vocal folds and a decrease of the pitch of voice.

Clinical notes

The **laryngeal inlet** and **rima glottidis** are narrow sections where foreign bodies may get stuck resulting in coughing and/or choking.

Laryngospasm is a spasmodic muscular contraction caused by irritation of the laryngeal mucosa closing the rima glottidis. This can result from trauma, foreign object aspiration, inflammation or allergy.

Laryngoscopy is an examination of the laryngeal cavity. It can be performed directly with a laryngoscope or indirectly using a mirror.

Cricothyrotomy (coniotomy) is performed by a transverse incision through the middle cricothyroid ligament and the conus elasticus.

Needle cricothyrotomy (coniotomy) is performed by inserting a needle into the laryngeal cavity through the cricothyroid ligament and the conus elasticus.

Both **cricothyrotomy** and **needle cricothyrotomy** are life-saving procedures.

Blood supply

Arterial supply:

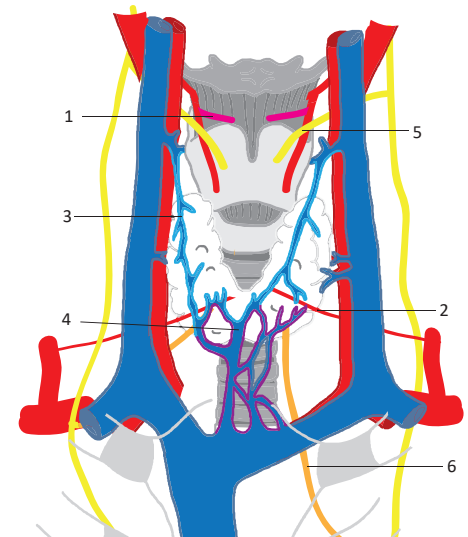
- 1 **Superior laryngeal artery** – superior thyroid artery (external carotid artery)
- 2 **Inferior laryngeal artery** – inferior thyroid artery (subclavian artery)

Venous drainage:

- 3 **Superior and middle thyroid veins** – internal jugular vein
- 4 **Inferior thyroid vein** – unpaired thyroid plexus (left brachiocephalic vein)

Lymphatic drainage:

– deep cervical lymph nodes and paratracheal lymph nodes



Blood supply and innervation of the larynx

Innervation

Viscerosensory innervation (innervating the mucosa): vagus nerve

- 5 **Superior laryngeal nerve** – the internal branch innervates the mucosa above the vocal folds
- 6 **Recurrent laryngeal nerve** – the mucosa below the vocal folds

Somatomotor innervation (innervating the muscles of the larynx): vagus nerve

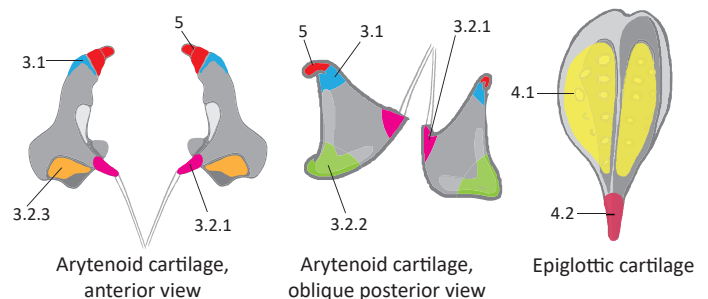
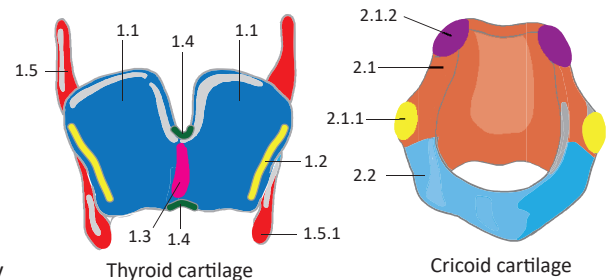
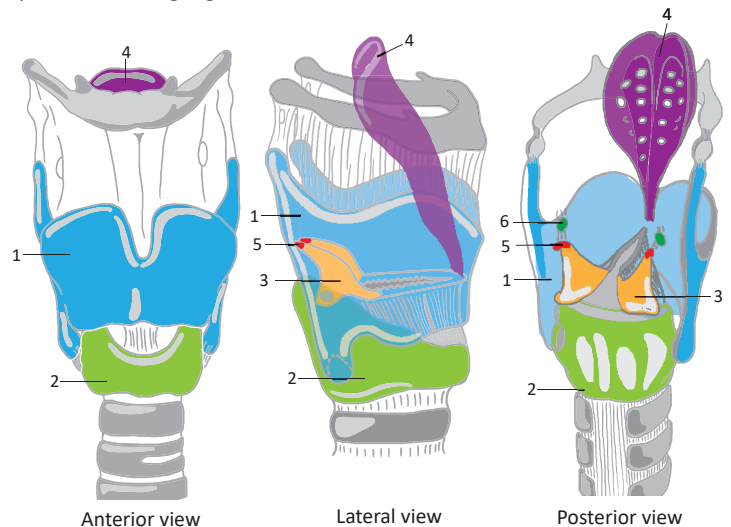
- 5 **Superior laryngeal nerve** – the external branch innervates the cricothyroid
- 6 **Recurrent laryngeal nerve** – innervates all the other laryngeal muscles

Parasympathetic innervation: vagus nerve

Sympathetic innervation: laryngopharyngeal branches from the superior cervical ganglion

Cartilages

- 1 **Thyroid cartilage** (*cartilago thyroidea*) – hyaline, unpaired
 - 1.1 **Left and right lamina** (*lamina dextra et sinistra*)
 - 1.2 **Oblique line** (*linea obliqua*)
 - 1.3 **Laryngeal prominence** (*prominentia laryngea*)
 - 1.4 **Superior and inferior thyroid notch** (*incisura thyroidea superior et inferior*)
 - 1.5 **Superior and inferior horns** (*cornua superiora et inferiora*)
 - 1.5.1 **Cricoid articular surface** (*facies articularis cricoidea*) – on the inferior horn
- 2 **Cricoid cartilage** (*cartilago cricoidea*) – hyaline, unpaired – the only completely closed cartilage around the larynx
 - 2.1 **Lamina** – the posterior part
 - 2.1.1 **Thyroid articular surface** (*facies articularis thyroidea*) – on the lateral border of the lamina and arch
 - 2.1.2 **Arytenoid articular surface** (*facies articularis arytenoidea*) – on the cranial margin of the lamina
 - 2.2 **Arch** (*arcus*) – the ventral part of the cricoid cartilage
- 3 **Arytenoid cartilage** (*cartilago arytenoidea*) – hyaline, paired
 - 3.1 **Apex** – the cranial elastic narrower part
 - 3.2 **Base** (*basis*) – the caudal hyaline wider part
 - 3.2.1 **Vocal process** (*processus vocalis*) – points ventromedially
 - 3.2.2 **Muscular process** (*processus muscularis*) – points dorsolaterally
 - 3.2.3 **Cricoid articular surface** (*facies articularis cricoidea*) – concave
- 4 **Epiglottic cartilage** (*cartilago epiglottica*) – cartilage of the epiglottis – elastic, unpaired
 - 4.1 **Lamina** – protrudes dorsocranially into the pharynx
 - 4.2 **Stalk of epiglottis** (*petiolus*) – is caudally attached to the internal side of the thyroid cartilage
- 5 **Corniculate cartilage** (*cartilago corniculata*) – elastic, paired – attaches to the apex of the arytenoid cartilage – forms the corniculate tubercle
- 6 **Cuneiform cartilage** (*cartilago cuneiformis*) – elastic, paired – is located laterocranially to the corniculate cartilage – is embedded in the ary-epiglottic fold – forms the cuneiform tubercle



Ligaments

Extrinsic membranes (true ligaments)

– connect the laryngeal cartilages together and the larynx to the surrounding structures

- **1 Thyrohyoid membrane** (*membrana thyrohyoidea*)
 - **1.1 Median thyrohyoid ligament** (*ligamentum thyrohyoideum medianum*)
 - **1.2 Lateral thyrohyoid ligament** (*ligamentum thyrohyoideum laterale*)
- **2 Cricothyroid ligament** (*elastic membrane / ligamentum cricothyroideum*)
 - extends from the caudal margin of the thyroid cartilage to the cranial margin of the cricoid cartilage
 - keeps the conus elasticus in the midline, located ventrally to it
- **3 Cricotracheal ligament** (*ligamentum cricotracheale*)
- 4 Thyro-epiglottic ligament** (*ligamentum thyroepiglotticum*)
 - connects the stalk of the epiglottis to the internal surface of the thyroid cartilage
- **5 Hyo-epiglottic ligament** (*ligamentum hyoepiglotticum*)

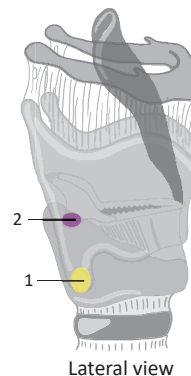
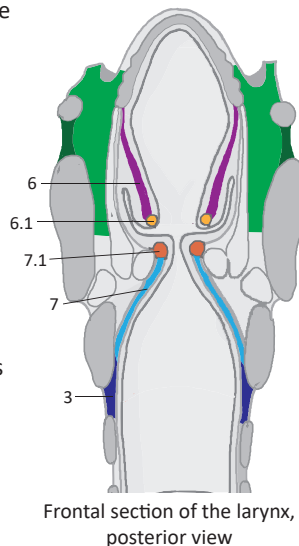
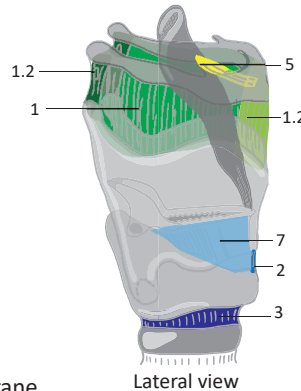
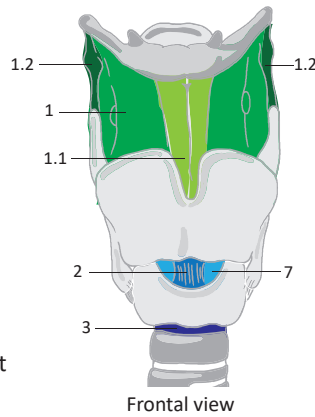
Intrinsic membranes – fibro-elastic laryngeal membranes (*membrana fibroelastica laryngis*)

– form a thickened fibrous layer under the submucosa

- **6 Quadrangular membrane** (*membrana quadrangularis*)
 - extends from the vestibular fold up to the lateral margin of the epiglottic cartilage
 - its cranial margin underlies the ary-epiglottic fold
 - **6.1 Vestibular ligament** (*ligamentum vestibulare*)
 - the thickened caudal margin of the quadrangular membrane extending from the dorsal surface of the thyroid cartilage to the ventrolateral surface of the arytenoid cartilage
 - the false ligament underlying the vestibular fold
- **7 Conus elasticus**
 - extends from the vocal folds to the internal circumference of the cricoid cartilage
 - is weakened in the midline
 - is continuously connected to the cricothyroid ligament
 - **7.1 Vocal ligament** (*ligamentum vocale*)
 - an elastic ligament forming the cranial margin of the conus elasticus
 - extends from the middle and dorsal parts of the thyroid cartilage to the dorsal part of the vocal process

Joints

- **1 Cricothyroid joint** (*articulatio cricothyroidea*)
 - Type:** simple
 - Shape:** cylindrical
 - Articular head:** cricoid articular surface on the inferior horn of thyroid cartilage
 - Articular fossa:** thyroid articular surface of cricoid cartilage
 - Movement:** ventrodorsal swaying movements
- **2 Crico-arytenoid joint** (*articulatio cricoarytenoidea*)
 - Type:** simple
 - Shape:** flattened elliptical
 - Articular head:** arytenoid articular surface of cricoid cartilage
 - Articular fossa:** cricoid articular surface of arytenoid cartilage
 - Movement:** shifts and rotations, which are fundamental for abduction and adduction of the vocal ligaments



Phonation (voice production) is made possible by tensing the vocal folds and narrowing the rima glottidis.

Ventilation can only occur when the vocal folds are relaxed and the rima glottidis is open.

A **cough** (*tussis*) is an expiratory reflex to clean the lower airways of phlegm, sputum and foreign bodies irritating or blocking the airways. A cough begins with a deep breath, followed by strong contraction of the abdominal and expiratory muscles against a closed rima glottidis. The rima glottidis suddenly opens and air is forced out.

The **tussigenic zones** consist of areas innervated by the vagus nerve that may induce coughing when irritated. They include the larynx, trachea, main bronchi, parietal pleura, diaphragm, pericardium, oesophagus and external acoustic meatus.

The **mechanism of sneezing** is very similar to coughing. The difference is that during a sneeze, the uvula is depressed and most of the air is forced out through the nose. The fifth cranial nerve innervating the nasal mucosa provides the afferent limb of the sneeze reflex.

During swallowing the airways are closed by elevation of the pharynx by the suprahyoid muscles and the thyrohyoid with subsequent closure of the laryngeal inlet by the epiglottis.

Closure of the laryngeal inlet is caused by contraction of the ary-epiglottic part of the oblique arytenoid, the thyro-epiglottic part of the thyro-arytenoid and by descent of the root of the tongue.

Closure of the rima glottidis is caused by contraction of the lateral crico-arytenoid and the oblique and transverse arytenoids.

Clinical notes

Injury of the recurrent laryngeal nerve occurs as a complication of thyroid surgery due to its close topographical relation to the thyroid gland. Unilateral injury may result in a quiet voice. Bilateral injury causes difficulty in breathing as a result of paralysis of the posterior cricoarytenoid.

Palsy of the posterior cricoarytenoid prevents patency through the rima glottidis during inspiration.

Laryngeal oedema (*oedema glottidis/laryngis*) is inflammation located under the mucosa of the laryngeal cavity.

Intrinsic laryngeal muscles

Ventral group

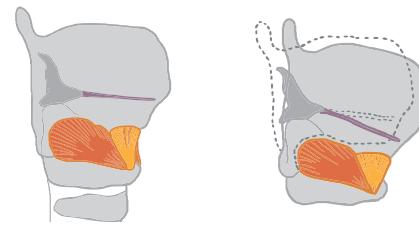
1 Cricothyroid (*musculus cricothyroideus*)

parts:

- straight part (*pars recta*)
- oblique part (*pars obliqua*)

course: from the ventral side of the cricoid cartilage laterocranially to the caudal margin of the thyroid cartilage

function: moves the thyroid cartilage ventrocaudally while concurrently moving the arch of the cricoid cartilage cranially, tightening the vocal folds



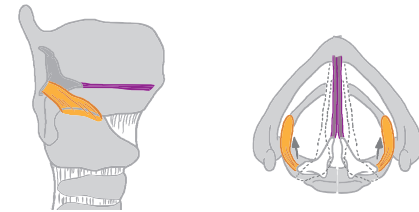
Cricothyroid

Lateral group

2 Lateral cricoarytenoid (*musculus cricoarytenoideus lateralis*)

course: from the cranial part of the lateral margin of the cricoid cartilage to the muscular process of the arytenoid cartilage

function: moves the muscular process ventrolaterally, adducting the vocal cords and closing the intermembranous part of the rima glottidis



Lateral cricoarytenoid

3 Thyroarytenoid (*musculus thyroarytenoideus*)

course: from the internal side of the thyroid cartilage along the external side of the vocal ligament to the vocal process of the arytenoid cartilage

function: internal rotation and ventral shift of the arytenoid cartilage – adduction and shortening of the vocal ligament (narrows the rima glottidis)

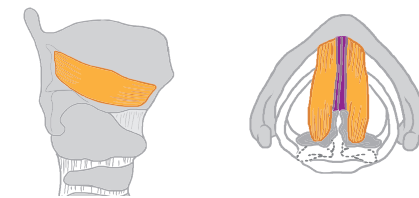
3.1 Vocalis (*musculus vocalis*)

- the medial part of the thyroarytenoid adjacent to the vocal ligament
- shortens and relaxes the vocal ligament
- the main muscle regulating strain on the vocal ligaments
- contains many motor units

3.2 Thyro-epiglottic part (*pars thyroepiglottica*)

course: fascicles of the thyroarytenoid from the thyroid cartilage extending to the lateral margin of the epiglottis

function: moves the margins of the epiglottis ventrally enlarging the rima glottidis



Thyroarytenoid



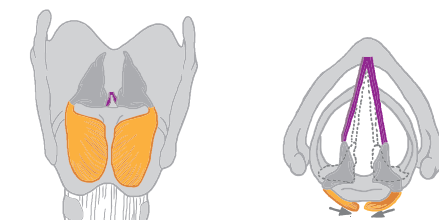
Thyro-epiglottic part of the thyroarytenoid

Dorsal group

4 Posterior cricoarytenoid (*musculus cricoarytenoideus posterior*)

course: begins on the dorsal side of the cricoid cartilage and runs craniolaterally to the muscular process of the arytenoid cartilage

function: external rotation and abduction of the arytenoid cartilage – abducts and lengthens the vocal ligament, enlarging the laryngeal inlet

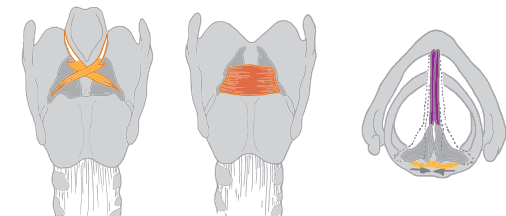


Posterior cricoarytenoid

5 Transverse arytenoid (*musculus arytenoideus transversus*)

course: transversally connects both arytenoid cartilages on their dorsal sides

function: adduction of the arytenoid cartilages – adducts the vocal ligaments, narrowing the rima glottidis (both the intercartilaginous and interarytenoid parts)



Transverse and oblique arytenoid

6 Oblique arytenoid (*musculus arytenoideus obliquus*)

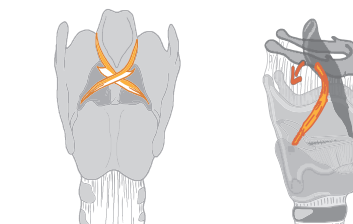
course: obliquely connects both arytenoid cartilages on their dorsal sides

function: adduction of the arytenoid cartilages – adducts the vocal ligaments, narrowing the rima glottidis (both the intercartilaginous and interarytenoid parts) – with the transverse arytenoid forms the strongest vocal ligament adductor

6.1 Ary-epiglottic part (*pars aryepiglottica*)

course: fascicles of the oblique arytenoid extending to the lateral margin of the epiglottic cartilage – forms the ary-epiglottic fold

function: moves the margins of the epiglottis dorsocaudally, and thus depresses the epiglottis, closing the laryngeal inlet

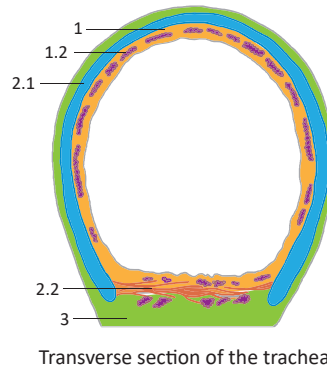


Ary-epiglottic part of the oblique arytenoid

The trachea is **connected to the larynx at the level of the vertebra C6**. It passes through the superior mediastinum and **bifurcates into the two main bronchi at the level of the vertebra T4**. It is **composed of 15–20 hyaline, C-shaped dorsally opened cartilages**, which are interconnected by a fibromuscular membrane.

Internal structure

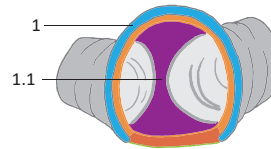
- **1 Mucosa and submucosa**
 - 1.1 Pseudostratified ciliated columnar epithelium
 - 1.2 Tracheal glands (*glandulae tracheales*)
 - seromucous glands
 - 1.3 Lymphoid tissue
- 2 Fibromusculocartilaginous layer (*tunica fibromusculocartilaginea*)
 - 2.1 Tracheal cartilage (*cartilago trachealis*)
 - 2.2 Membranous wall (*paries membranaceus*)
 - a fibromuscular membrane connecting the tracheal cartilages together at their dorsal edges
- 3 Adventitia (*tunica adventitia*)



Transverse section of the trachea

External structure

- **1 Tracheal cartilages** (*cartilagines tracheales*)
 - 15–20 hyaline cartilages
 - the last cartilage is triangular with a caudally pointing vertex
 - 1.1 Carina of trachea (*carina tracheae*)
 - the ridge of the last cartilage protruding into the lumen of the trachea
- 2 Annular ligaments (*ligamenta anularia*)
 - connect adjacent cartilages to each other



Superior view of the bifurcation of trachea

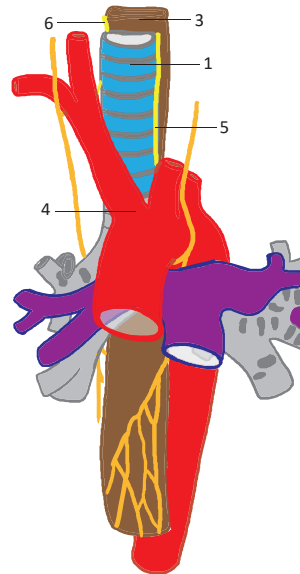
Syntopy

Cervical part (*pars cervicalis*)

- the continuation of the larynx at the level of the vertebra C6
- is located within the visceral space of the neck
- ventrally:** isthmus of the thyroid gland
- laterally:** thyroid gland, carotid sheath and recurrent laryngeal nerve (dorsolaterally)
- dorsally:** oesophagus

Thoracic part (*pars thoracica*)

- begins at the level of the cranial margin of the manubrium of the sternum
- passes through the superior mediastinum
- splits into the two main bronchi at the level of the T4 vertebra
- near the bifurcation, its left side touches the arch of the aorta
- **3 dorsally:** oesophagus
- **4 ventrally:** arch of aorta and its branches
- **5 left:** left recurrent laryngeal nerve, arch of aorta
- **6 right:** right recurrent laryngeal nerve, superior vena cava and azygos vein



Syntopy of the trachea, anterior view

Blood supply

Arterial supply: tracheal branches (inferior thyroid artery), bronchial branches (descending aorta)

Venous drainage: tracheal veins (to the veins of the thyroid gland and oesophagus)

Lymphatic drainage: paratracheal nodes, tracheobronchial nodes

Innervation

Sympathetic innervation: parietal plexuses from the stellate ganglion

Parasympathetic innervation: recurrent laryngeal nerve, tracheal branches (vagus nerve)

Viscerosensory innervation: recurrent laryngeal nerve, tracheal branches (vagus nerve)

The bronchopericardial membrane extends between the main bronchi, oesophagus and the dorsal side of the pericardium, and forms the borderline between the anterior and posterior mediastinum.

Proportions of the trachea:

length 10–11 cm,
diameter 14–18 mm.

Length of the main bronchi:

The right main bronchus is about 3 cm, while the left main bronchus is 4–5 cm.

The ventral surface of the oesophagus is in contact with the flexible membranous wall of the trachea (*paries membranaceus tracheae*). This allows ventral extension of the oesophageal wall during swallowing.

Clinical notes

Tracheotomy is a surgical procedure in which a horizontal or longitudinal incision through the ventral side of the trachea is performed. An upper tracheotomy (*tracheotomia superior*) is performed below the cricoid cartilage and above the isthmus of the thyroid gland. A lower tracheotomy (*tracheotomia inferior*) is performed below the isthmus of the thyroid gland at the level of the third tracheal cartilage. The resulting hole (stoma) of tracheotomy is called a tracheostomy.

A **congenital tracheo-oesophageal fistula** is a persisting communication between the respiratory and digestive tracts caused by incomplete separation of the oesophagus and the trachea.

Endotracheal intubation is the placement of a tube into the trachea through the oral cavity (orotracheal intubation) or the nasal cavity (nasotracheal intubation) to allow ventilation.

Postintubation tracheal stenosis occurs in 1 % of intubated patients. It usually becomes symptomatic after 3 months of intubation. Patients usually present with emphysema and inspiratory stridor. The stenosis can be treated by interventional bronchoscopy or by surgical resection of the trachea.

75 % of aspirated foreign bodies enter the right main bronchus due to its wider diameter and its direct continuity with the trachea.

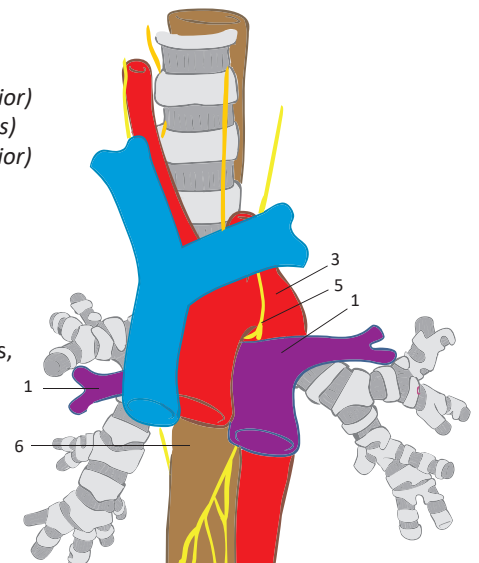
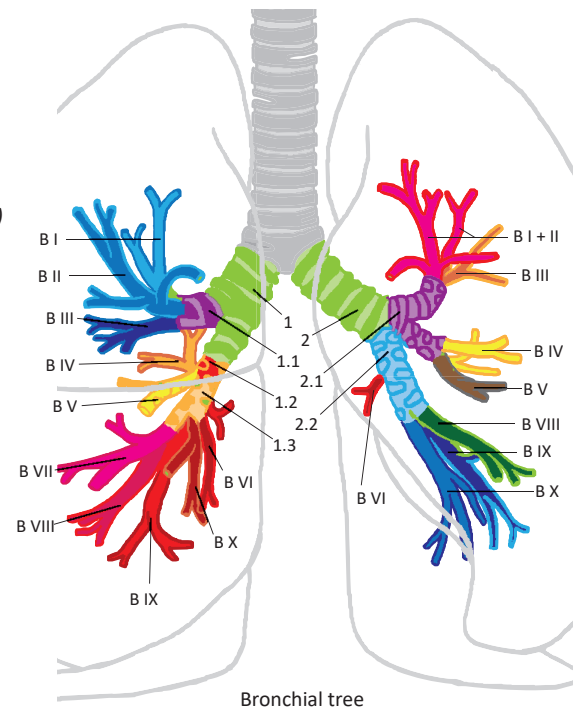
The bronchi arise from the tracheal bifurcation at the level of the sternal angle and successively divide to form the bronchial tree. The main bronchi run in the mediastinum and divide into the lobar bronchi in the hilum of the lung. Peripheral parts of the bronchial tree are located within the lung tissue itself. Cartilaginous rings are incomplete, dorsally opened C-shaped rings in the primary bronchi. They become irregular and infrequent towards the periphery of the bronchial tree.

Basic bronchial tree

1. **Main/primary bronchi** (*bronchi principales*) – for the right and left lung
2. **Lobar/secondary bronchi** (*bronchi lobares*) – for the three lobes of the right lung and the two lobes of the left lung
3. **Segmental/tertiary bronchi** (*bronchi segmentales*) – for the 10 segments of the right lung and the (usually) 8 segments of the left lung

Bronchial tree (*arbor bronchialis*)

- 1 **Right main (primary) bronchus** (*bronchus principalis dexter*) – shorter, wider, more vertical and less deviated from the tracheal axis
 - 1.1 **Right superior lobar bronchus** (*bronchus lobaris superior dexter*)
 - B I **Apical segmental bronchus** (*bronchus segmentalis apicalis*)
 - B II **Posterior segmental bronchus** (*bronchus segmentalis posterior*)
 - B III **Anterior segmental bronchus** (*bronchus segmentalis anterior*)
 - 1.2 **Middle lobar bronchus** (*bronchus lobaris medius*)
 - B IV **Lateral segmental bronchus** (*bronchus segmentalis lateralis*)
 - B V **Medial segmental bronchus** (*bronchus segmentalis medialis*)
 - 1.3 **Right inferior lobar bronchus** (*bronchus lobaris inferior dexter*)
 - B VI **Superior segmental bronchus** (*bronchus segmentalis superior*)
 - B VII **Medial basal segmental bronchus** (*bronchus segmentalis basalis medialis or cardiacus*)
 - B VIII **Anterior basal segmental bronchus** (*bronchus segmentalis basalis anterior*)
 - B IX **Lateral basal segmental bronchus** (*bronchus segmentalis basalis lateralis*)
 - B X **Posterior basal segmental bronchus** (*bronchus segmentalis basalis posterior*)
- 2 **Left main (primary) bronchus** (*bronchus principalis sinister*) – longer, narrower, laterocranially concave and more deviated from the tracheal axis
 - 2.1 **Left superior lobar bronchus** (*bronchus lobaris superior sinister*)
 - B I + II **Apicoposterior segmental bronchus** (*bronchus segmentalis apicoposterior*)
 - B III **Anterior segmental bronchus** (*bronchus segmentalis anterior*)
 - B IV **Superior lingular bronchus** (*bronchus lingularis superior*)
 - B V **Inferior lingular bronchus** (*bronchus lingularis inferior*)
 - 2.2 **Left inferior lobar bronchus** (*bronchus lobaris inferior sinister*)
 - B VI **Superior segmental bronchus** (*bronchus segmentalis superior*)
 - B VII **Medial basal segmental bronchus** – missing in 90 % of cases (*bronchus segmentalis basalis medialis or cardiacus*)
 - B VIII **Anterior basal segmental bronchus** (*bronchus segmentalis basalis anterior*)
 - B IX **Lateral basal segmental bronchus** (*bronchus segmentalis basalis lateralis*)
 - B X **Posterior basal segmental bronchus** (*bronchus segmentalis basalis posterior*)



Synty

- 1 **Pulmonary arteries** (*arteriae pulmonales*) – ventrocaudal to the main right bronchus and cranial to the main left bronchus
- 2 **Pulmonary veins** (*venae pulmonales*) – ventrocaudal to the main bronchi
- 3 **Aortic arch** (*arcus aortae*) – superiorly convex, starts anteriorly to the left main bronchus, as it curves it becomes superior, and finally posterior to the left main bronchus
- 4 **Azygos vein** (*vena azygos*) – ascends ventrally to the right main bronchus, arches over it and opens into the superior vena cava
- 5 **Vagus nerves** (*nervus vagus*) – dorsal to the main bronchi
- 6 **Oesophagus** – dorsal to the left main bronchus

Blood supply and innervation

Arterial supply: bronchial branches (thoracic aorta, posterior intercostal arteries)

Venous drainage: bronchial veins (azygos vein, accessory hemiazygos vein, posterior intercostal veins)

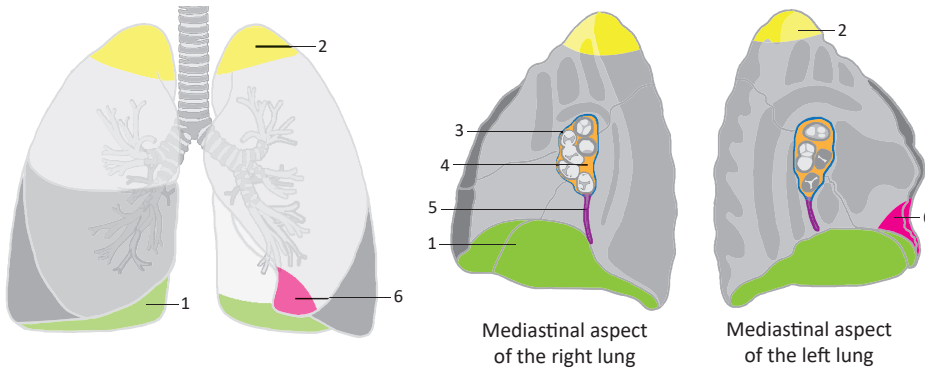
Lymphatic drainage: intrapulmonary nodes (further to the bronchopulmonary nodes and then to the tracheobronchial nodes)

Mixed innervation: peribronchial plexus (viscerosensory; visceromotor: sympathetic, parasympathetic – vagus nerve)

The lung is a **paired organ covered with the pleura** and located in the **pleural cavity**. Pulmonary tissue consists of the bronchial tree, connective tissue (the pulmonary interstitium), vessels, nerves and lymphatic nodes. The lungs are divided into **lobes each with its own lobar bronchus**. The lobes are further divided into segments, defined as the area ventilated by a segmental bronchus and perfused by a branch of the pulmonary artery. The **right lung has 3 lobes** and 10 segments; the left lung has **2 lobes** and usually 8 segments.

Basic parts

- 1 **Base of lung** (*basis pulmonis*) – the concave base resting on the diaphragm
- 2 **Apex of lung** (*apex pulmonis*) – projects towards the cervical pleura
- 3 **Hilum of lung** (*hilum pulmonis*) – a place where the structures of the root of the lung enter the lung
- 4 **Root of lung** (*radix pulmonis*) – the vessels, lymphatics, nerves, and bronchi passing through the hilum
- 5 **Pulmonary ligament** (*ligamentum pulmonale*) – a caudally extending pleural fold – a transition between the parietal and visceral pleura
- 6 **Lingula of left lung** (*lingula pulmonis sinistri*) – a thin slip of pulmonary tissue extending from the upper lobe of the left lung – the analogue of the middle lobe on the right lung



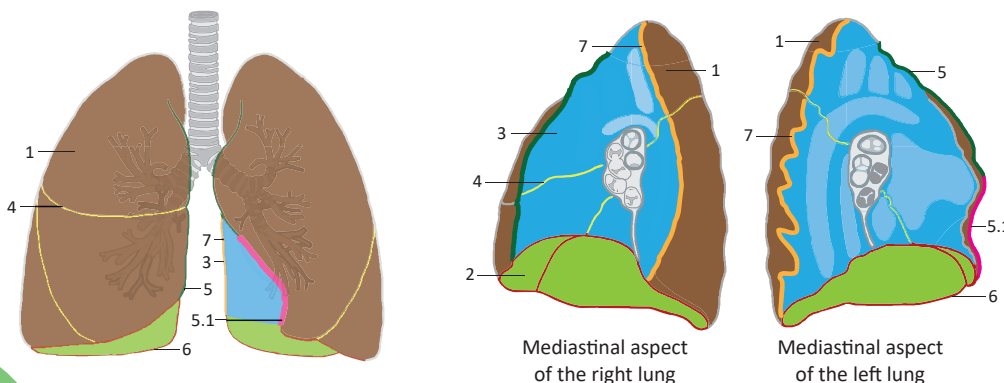
Surfaces and borders

Surfaces

- 1 **Costal surface** (*facies costalis*) – faces the ribs
- 2 **Diaphragmatic surface** (*facies diaphragmatica*) – faces the diaphragm
- 3 **Medial surface** (*facies medialis*) – faces the mediastinum (mediastinal part) and the vertebral column (vertebral part)
- 4 **Interlobar surfaces** (*facies interlobares*) – formed by grooves between the lobes – face neighbouring lobes and are covered by the visceral pleura

Borders

- 5 **Anterior border** (*margo anterior*) – a sharp edge between the costal and medial surfaces
 - 5.1 **Cardiac notch of left lung** (*incisura cardiaca*) – a deep indentation on the anterior border of the left lung for the heart
- 6 **Inferior border** (*margo inferior*) – runs between the diaphragmatic and costal surfaces
- 7 **Vertebral border** (*margo vertebralis*) – the dorsal transition between the costal and medial surfaces



The **pulmonary interstitium** is the connective tissue supporting the alveoli. It can be divided into:

1. peribronchial and perihilar fibrous tissue
2. interalveolar (or more simply alveolar), interlobular and intersegmental septa.

Surfactant is a superficial layer of proteins and phospholipids lining the alveoli. It reduces surface tension, which prevents the alveoli from collapsing. It is an acronym from **SUR**face **ACT**ive **AgE**NT.

The **airways of the lungs** ramify as the bronchial tree (first 14 levels), which serves for the air conduction, and then as the alveolar tree (other 9 levels) which serves for the air conduction and respiration.

The **medial basal segment S VII** (*segmentum basale mediale*) of the left lung is present in approximately 10 % of cases. In the other cases, it is incorporated into S VIII.

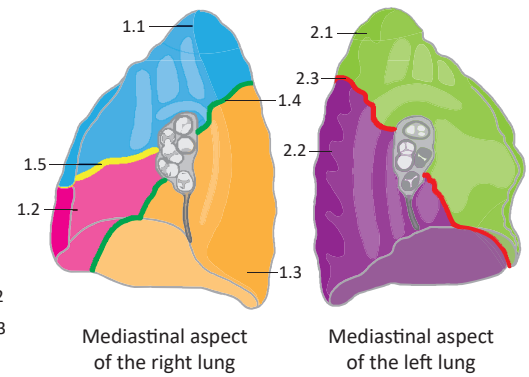
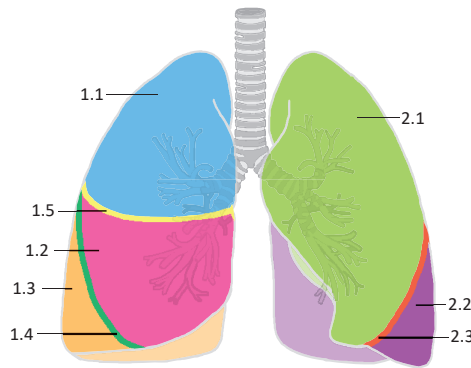
Pulmonary lobes (*lobi pulmonis*)

1 Right lung (*pulmo dexter*)

- 1.1 Superior lobe (*lobus superior*)
- 1.2 Middle lobe (*lobus medius*)
- 1.3 Inferior lobe (*lobus inferior*)
- 1.4 Oblique fissure (*fissura obliqua*)
- 1.5 Horizontal fissure (*fissura horizontalis*)

2 Left lung (*pulmo sinister*)

- 2.1 Superior lobe (*lobus superior*)
- 2.2 Inferior lobe (*lobus inferior*)
- 2.3 Oblique fissure (*fissura obliqua*)



Bronchopulmonary segments (*segmenta bronchopulmonalia*)

Right lung (*pulmo dexter*)

Superior lobe (*lobus superior*)

- S I Apical segment (*segmentum apicale*)
- S II Posterior segment (*segmentum posterius*)
- S III Anterior segment (*segmentum anterius*)

Middle lobe (*lobus medius*)

- S IV Lateral segment (*segmentum laterale*)
- S V Medial segment (*segmentum mediale*)

Inferior lobe (*lobus inferior*)

- S VI Superior segment (*segmentum superius*)
- S VII Medial basal segment (*segmentum basale mediale / segmentum cardiacum*)
- S VIII Anterior basal segment (*segmentum basale anterius*)
- S IX Lateral basal segment (*segmentum basale laterale*)
- S X Posterior basal segment (*segmentum basale posterius*)

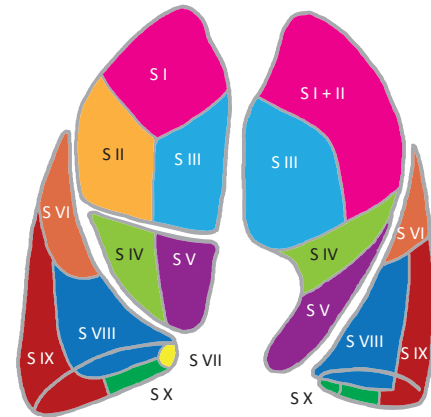
Left lung (*pulmo sinister*)

Superior lobe (*lobus superior*)

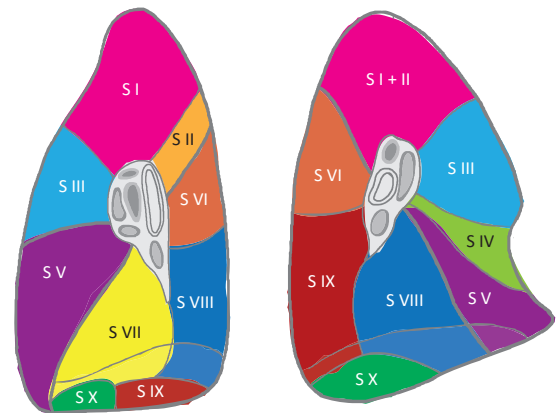
- S I + II Apicoposterior segment (*segmentum apicoposterius*)
- S III Anterior segment (*segmentum anterius*)
- S IV Superior lingular segment (*segmentum lingulare superius*)
- S V Inferior lingular segment (*segmentum lingulare inferius*)

Inferior lobe (*lobus inferior*)

- S VI Superior segment (*segmentum superius*)
- S VIII Anterior basal segment (*segmentum basale anterius*)
- S IX Lateral basal segment (*segmentum basale laterale*)
- S X Posterior basal segment (*segmentum basale posterius*)



Right and left lung, anterior view

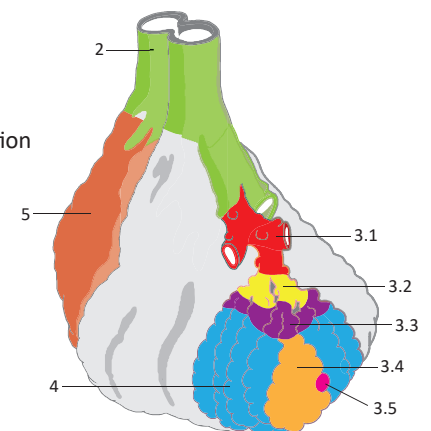


Mediastinal aspect of the right lung

Mediastinal aspect of the left lung

Subsegments of lung parenchyma

- 1 Bronchioles (*bronchioli*) – contain neither cartilage nor glands
- 2 Terminal bronchioles (*bronchioli terminales*) – the epithelium changes to simple columnar epithelium with exocrine bronchiolar cells (Clara cells)
- 3 Alveolar tree (*arbor alveolaris*) – the peripheral respiratory part of the bronchial segmentation
 - 3.1 Respiratory bronchioles (*bronchioli respiratorii*) – first alveoli occur
 - 3.2 Alveolar ducts (*ductus alveolares*) – contain many alveoli
 - 3.3 Alveolar atria (*atria alveolaria*) – the widened terminal parts of the alveolar ducts
 - 3.4 Alveolar sacculles / alveolar sacs (*sacculi alveolares*) – the end of the bronchial tree
 - 3.5 Pulmonary alveoli (*alveoli pulmonis*) – blind ends (cul-de-sacs) of the bronchial tree proceeding from the respiratory bronchioles, alveolar ducts and alveolar sacculles
- 4 Primary pulmonary lobule (*lobulus pulmonis primarius*) – is ventilated by one respiratory bronchiole
- 5 Secondary pulmonary lobule (*lobulus pulmonis secundarius*) – the basic structural and functional unit of pulmonary tissue – is ventilated by one terminal bronchiole



Subsegments of the lung parenchyma

Syntopy

1 Mediastinal surface (*facies mediastinalis*)

On both lungs

- 1.1 Groove for first rib (*sulcus costae primae*)
- 1.2 Groove for subclavian artery (*sulcus arteriae subclaviae*)
- 1.3 Cardiac impression (*impressio cardiaca*)

On the right lung

- 1.4 Groove for superior vena cava (*sulcus venae cavae superioris*)
- 1.5 Groove for oesophagus (*sulcus oesophageus*)
- 1.6 Groove for azygos vein (*sulcus venae azygos*)

On the left lung

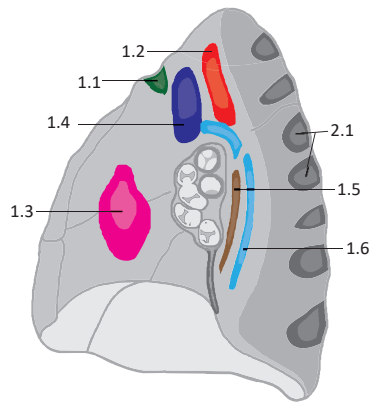
- 1.7 Groove for brachiocephalic vein (*sulcus venae brachiocephalicae sinistrae*)
- 1.8 Groove for aorta (*sulcus aorticus*)
- 1.9 Oesophageal impression (*impressio oesophagea*)

2 Costal surface (*facies costalis*)

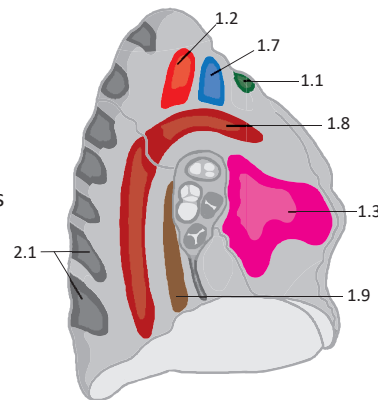
- 2.1 Costal impressions (*impressiones costarum*)

3 Diaphragmatic surface (*facies diaphragmatica*)

- is in direct contact with the diaphragm
- the following three organs are in contact with the lungs through the diaphragm (but do not form impressions):
- 3.1 Liver – in contact with the right lung
- 3.2 Stomach – in contact with the left lung
- 3.3 Spleen – in contact with the left lung



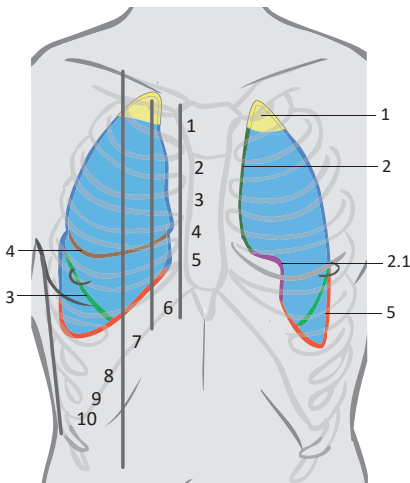
Mediastinal aspect of the right lung



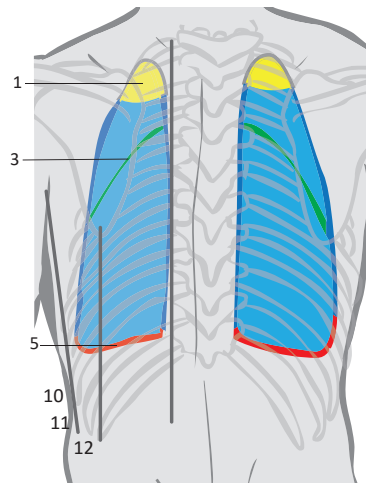
Mediastinal aspect of the left lung

Borders and projection of the lungs

- 1 Apex of lung (*apex pulmonis*)
 - is covered by the pleural cupula in the lesser supraclavicular fossa
- 2 Anterior border (*margo anterior*)
 - retrosternally between the anterior median line and the sternum
 - cranially recedes from the anterior median line to the lung apex
 - caudally recedes from the anterior median line
 - 2.1 Cardiac notch (*incisura cardiaca*)
 - the curved margin around the pericardium on the left lung
 - extends from the 4th sternocostal joint to the 6th rib in the midclavicular line
- 3 Oblique fissure (*fissura obliqua*) – extends from the spinous process T3 along the 6th rib to the navel
- 4 Horizontal fissure (*fissura horizontalis*) – on the right side from the oblique fissure in the midaxillary line along the 4th rib to the sternum
- 5 Inferior border (*margo inferior*)
 - at the level of the 6th rib in the midclavicular line
 - at the level of the 8th rib in the midaxillary line
 - at the level of the 10th rib in the scapular line



Anterior view



Posterior view

A mnemonic for remembering the arrangement of the structures in the hilum of the lung (Bronchus, Artery, Vein in a craniocaudal direction):

BRIGHT IS RIGHT – bronchus on the right is cranial – but not on the left side

A mnemonic for remembering the position of the vagus and phrenic nerves in relation to the hilum:

Vagus nerve – behind the hilum of the lung (“vagus stays in the back”)

Phrenic nerve – in front of the hilum of the lung (“phrenic is in the front”)

There are many **anastomoses** between the pulmonary and bronchial vascular beds (arterio-arterial and veno-venous) as well as between the arteries and veins in the pulmonary bed (arterio-venous). These connections may function as right-left shunts.

Nociceptive neurons (neurons perceiving the pain) are present in small numbers in the lungs and visceral pleura (lungs and visceral pleura do not have pain sensation). The only exception is the area around the hilum of the lung supplied by the vagus nerve.

Intrapulmonary lymph nodes are located around the branching points of the segmental bronchi in the lungs.

Bronchopulmonary lymph nodes are located around the lobar bronchi in the hilum of the lung.

Tracheobronchial lymph nodes are located around the tracheal bifurcation

Clinical notes

Pulmonary embolism is a life-threatening condition, which occurs due to obstruction of the pulmonary artery or any of its branches. In most cases it is due to a thromboembolism originating in the pelvic veins or the deep veins of the leg. Infrequently other types of embolism may occur, such as fat embolism (from fractures), amniotic fluid embolism (during delivery), air embolism (during injury) and tumour cell embolism.

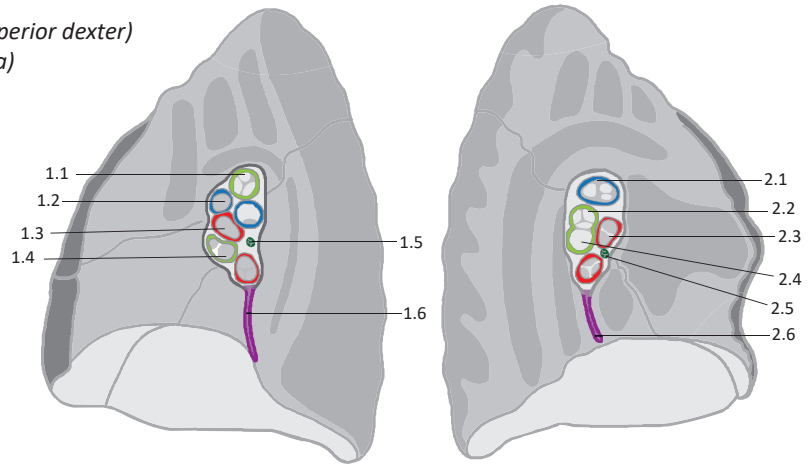
Structures in the hilum of the lung

1 Hilum of right lung (*hilum pulmonis dextrum*)

- 1.1 Right superior lobar bronchus (*bronchus lobaris superior dexter*)
- 1.2 Right pulmonary arteries (*arteria pulmonalis dextra*)
- 1.3 Right pulmonary veins (*venae pulmonales dextrae*)
- 1.4 Middle and inferior lobar bronchi (*bronchus lobaris medius et inferior dexter*)
- 1.5 Right bronchopulmonary lymph nodes (*nodi lymphoidei bronchopulmonales dextri*)

2 Hilum of left lung (*hilum pulmonis sinistrum*)

- 2.1 Left pulmonary artery (*arteria pulmonalis sinistra*)
- 2.2 Left superior lobar bronchus (*bronchus lobaris superior sinister*)
- 2.3 Left pulmonary veins (*venae pulmonales sinistrae*)
- 2.4 Left inferior lobar bronchus (*bronchus lobaris inferior sinister*)
- 2.5 Left bronchopulmonary lymph nodes (*nodi lymphoidei bronchopulmonales sinistri*)
- 2.6 Pulmonary ligament (*ligamentum pulmonale*)



Mediastinal aspect of the right lung

Mediastinal aspect of the left lung

Blood supply

Functional circulation (*vasa publica*) – branches of the pulmonary artery and tributaries of the pulmonary veins

1. **Pulmonary trunk (*truncus pulmonalis*)** – begins at the right ventricle of the heart (contains deoxygenated blood)
2. **Right and left pulmonary arteries (*arteria pulmonalis dextra et sinistra*)** – branch along the bronchial tree
3. **Veins** – begin in the interstitial connective tissue and travel independently of the bronchial tree (the larger veins travel with the bronchi)
4. **Pulmonary veins (*venae pulmonales*)** – two from the right lung and two from the left lung – empty into the left atrium of the heart (contain oxygenated blood)

Nutritive circulation (*vasa privata*)

1. **Bronchial branches (*rami bronchiales*)** – contain oxygenated blood and course along the bronchial tree – usually two on the left from the thoracic aorta and one on the right from the third intercostal artery – form anastomoses with the bed of the pulmonary artery from the respiratory bronchioles onwards
2. **Bronchial veins (*venae bronchiales*)** – travel along the bronchial tree from the middle-calibre bronchi onwards
3. **To azygos vein, accessory hemi-azygos vein and intercostal veins**

Lymphatic drainage

- the superficial and deep lymphatic systems drain lymph to the hilar nodes
- from the hilar nodes lymph is drained through the bronchomediastinal trunks into the main lymphatic ducts

Superficial system

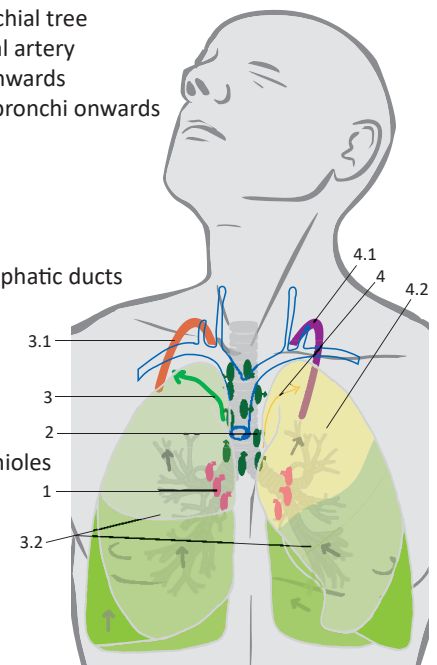
1. plexuses within the subpleural connective tissue
2. along the pulmonary veins (perivenous location)
3. there are no interposed lymph nodes before entering the bronchopulmonary nodes

Deep system

1. plexuses in the peribronchial connective tissue beginning at the level of the respiratory bronchioles
2. along the bronchi (peribronchial location)
3. intrapulmonary nodes are interposed before entering the bronchopulmonary nodes

Lymph nodes and lymphatic trunks

- 1 **Bronchopulmonary nodes (*nodi lymphoidei bronchopulmonales*)** – hilar nodes
- 2 **Right and left superior and inferior tracheobronchial nodes (*nodi lymphoidei tracheobronchiales superiores et inferiores dextri et sinistri*)**
- 3 **Right bronchomediastinal trunk (*truncus bronchomediastinalis dexter*)** – lymphatic collector
 - 3.1 to the right lymphatic duct (*ductus lymphaticus dexter*)
 - 3.2 **drainage area:** whole right lung, inferior lobe of left lung and both lingular segments of left lung superior lobe (S IV and S V)
- 4 **Left bronchomediastinal trunk (*truncus bronchomediastinalis sinister*)** – lymphatic collector
 - 4.1 to the thoracic duct (*ductus thoracicus*)
 - 4.2 **drainage area:** superior lobe of left lung (S I – III) excluding both lingular segments



Lymphatic drainage of the lungs

Innervation

Sympathetic innervation: thoracic pulmonary branches – in the pulmonary plexus and the periarterial and peribronchial plexuses

Parasympathetic innervation: vagus nerve – in the pulmonary plexus and also in the periarterial and peribronchial plexuses

Viscerosensory innervation: by both sympathetic and parasympathetic nerves

The pleura is a **smooth glossy membrane** lining the thoracic cavity. It consists of a **layer of loose subserous tissue** covered by the **mesothelium**. It is divided into the **visceral pleura**, which is firmly **attached to the surface of the lungs** and extends into the interlobar fissures, and the **parietal pleura**, which **covers the inner surface of the thoracic cavity**. The **pleural cavity** is located **between these two layers**; it contains a small amount of **pleural fluid** (approximately 15 ml). This permits **smooth movements** of the lungs and pleura **during breathing** and **ensures negative pressure** in the thoracic cavity, keeping the lungs under tension.

Basic parts

- 1 **Visceral pleura** (*pleura visceralis*)
- 2 **Mesopneumonium** – the transition between the visceral and parietal pleura in the hilum of the lung
- 3 **Pleural cavity** (*cavitas pleuralis*)
- 4 **Parietal pleura** (*pleura parietalis*)
 - 4.1 **Pleural cupula** (*cupula pleurae*)
 - is located cranially between the scapuli
 - 4.2 **Costal part** (*pars costalis*) – faces the ribs
 - 4.3 **Mediastinal part** (*pars mediastinalis*) – faces mediastinum
 - 4.4 **Diaphragmatic part** (*pars diaphragmatica*)
 - faces the diaphragm

Pleural recesses (*recessus pleurales*)

- 5 **Costodiaphragmatic recess** (*recessus costodiaphragmaticus*)
 - between the ribs and the diaphragm
 - the deepest part of the pleural cavity
- 6 **Costomediastinal recess** (*recessus costomediastinalis*)
 - the anterior border of the mediastinal pleura between the ribs and the anterior mediastinum
- 7 **Phrenicosternal recess** (*recessus phrenicosternalis*)
 - between the diaphragm and the inferior mediastinum
- 8 **Vertebromediastinal recess** (*recessus vertebromediastinalis*)
 - between the vertebrae and the mediastinum

Pleural cupula / dome of pleura (*cupula pleurae*)

- 1 **Suprapleural membrane** – fascia of Sibson (*membrana suprapleuralis*)
 - the continuation of the endothoracic fascia
- 2 **Vertebropleural ligament** (*ligamentum vertebropleurale*)
 - from the deep layer of the cervical fascia
- 3 **Transversopleural ligament** (*ligamentum transversopleurale*)
 - from the transverse processes of the C6 and C7 vertebrae
- 4 **Costopleural ligament** (*ligamentum costopleurale*) – from the 1st rib
- 5 **Scalenopectoral ligament** (*ligamentum scalenopectoralis*)
 - from the inner surface of the scalenus anterior muscle

Blood supply

Arterial supply and venous drainage:

- **parietal pleura** – internal thoracic artery and veins, posterior intercostal veins and arteries
- **visceral pleura** – veins and arteries of the lungs

Innervation

Viscerosensory innervation (visceral pleura):

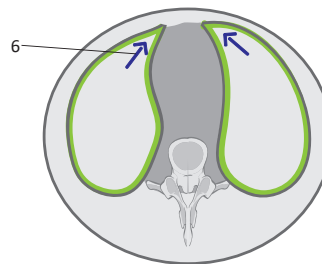
vagus nerve (only in the hilum of the lung)

Somatosensory innervation (parietal pleura):

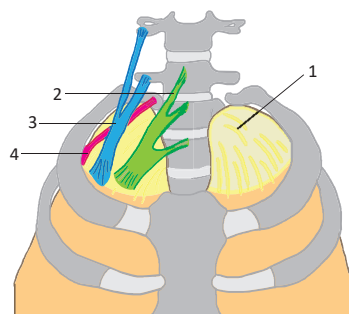
- **costal part, margins of the diaphragmatic part**: intercostal nerves
- **middle of the diaphragmatic part, mediastinal part**: phrenic nerve
- **pleural cupula**: brachial plexus



The schematic frontal section of the left pleural cavity



The schematic transversal section of the thorax



The components of the pleural cupula

Pleural villi are appendages on the visceral pleura around the margins of the pulmonary lobes. The lungs expand into the **recesses of the parietal pleura** during maximal inspiration.

The **parietal and visceral pleura** are attached to the surrounding structures by several interconnected layers:

- the fibrous layer of the pleura (subserosa, subserous layer)
- a layer of loose connective tissue under the pleura (subpleural layer)
- interstitium of the lung, endothoracic fascia and the interstitial connective tissue of the mediastinum

The **phrenicopleural fascia** is a solid adhesion of the endothoracic fascia on the upper surface of the diaphragm to a layer of the subpleural connective tissue and a fibrous layer of the diaphragmatic part of the parietal pleura. Due to this solid adhesion, this diaphragmatic part of the parietal pleura is inseparable from the diaphragm.

The **fascia of Sibson** is a continuation of the endothoracic fascia covering the pleural cupula. It extends from the transverse processes of the 6th and 7th cervical vertebrae to the 1st rib.

The **parietal pleura** contains many nociceptors (“parietal pleura hurts”). In contrast, the visceral pleura contains many mechanoreceptors (“visceral pleura and lungs do not hurt”).

Clinical notes

Pneumothorax is an accumulation of gas in the pleural cavity. The cause may be: injury to the thorax, rupture of a subpleural bulla, or iatrogenic (e.g. cannulation of the subclavian vein or puncture of the pleural cavity)

Fluidothorax is a collection of fluid in the pleural cavity. It may be classified as haemothorax (blood), chylothorax (lymph) or pyothorax (pus and bacteria). If the fluid contains a high content of protein, it is called an exudate. If it is poor in protein, it is called a transudate.

Thoracentesis (pleural tap) is a procedure to remove fluid or air from the pleural cavity. A tube is inserted through the 2nd or 3rd intercostal space in the midclavicular line (half-sitting position) for pneumothorax, or through the 6th intercostal space in the anterior or middle axillary line (sitting position) for fluidothorax. It is frequently performed under ultrasound guidance.

Borders and projections of the pleura

Upper border:

- **Lesser supraclavicular fossa** (*fossa supraclavicularis minor*)
– the location of the cervical pleura

Anterior borders:

- the superior interpleural area (*area interpleuralis superior/thymica*) is bounded by the sternoclavicular joint cranially and the 2nd sternocostal joints caudally
- from the 2nd to the 4th ribs dorsally and parallel with the sternum
- from the 4th rib the borders diverge
- then form the inferior interpleural area (*area interpleuralis inferior or pericardiaca*)
- on the right, the border descends vertically along the sternum to the level of the 6th rib
- on the left, the border turns laterally towards the cartilage of the 6th rib

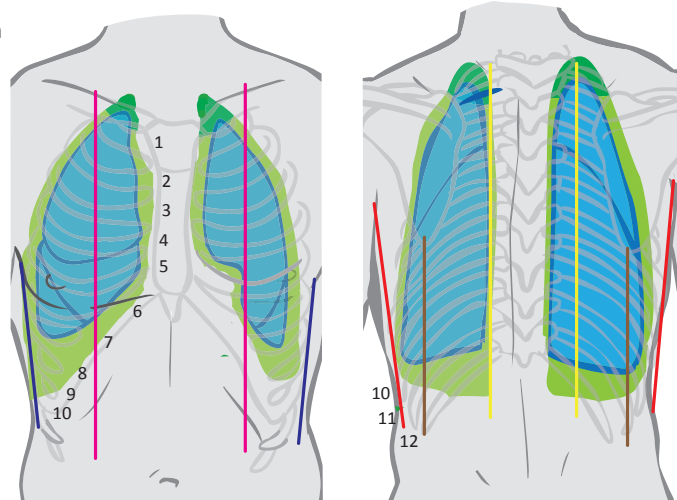
Inferior borders:

- from the level of the 6th rib the borders diverge and cross the anatomical lines
- 7th rib – midclavicular line
- 8th rib – anterior axillary line
- 9th rib – middle axillary line

Dorsal border:

- 10th rib – posterior axillary line
- 11th rib – scapular line
- 12th rib (reaches also 2 cm below its level) – paravertebral line
– the lowest and narrowest part of the pleural cavity

● Parietal pleura ● Visceral pleura



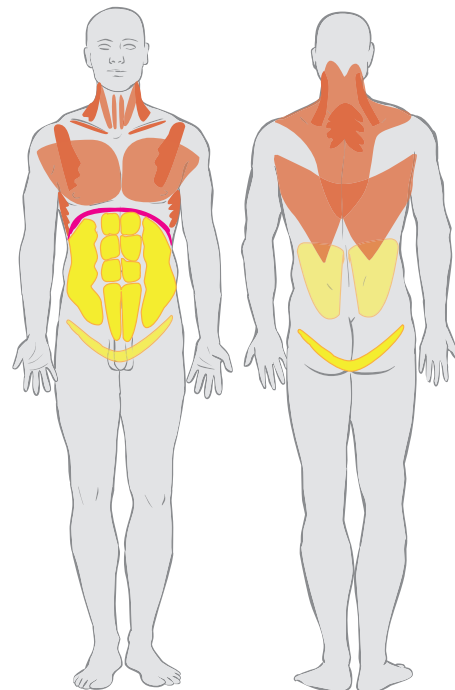
Anterior and posterior views of the parietal and visceral pleura

Mechanics of breathing

Inspiration is an **active process** in which the respiratory muscles elevate the thoracic cage ventrally and laterally. **Expiration** is a **passive process** in which the lung volume decreases due to the elastic recoil of the lungs. Forced expiration requires contraction of the expiratory muscles. **The inspiratory muscles insert on the superior parts of the ribs and elevate them. The expiratory muscles insert on the inferior parts of the ribs and depress them. The lungs are passive organs;** during breathing their volume adapts to the changes in volume and pressure in the thoracic cavity. Both the visceral and parietal pleura are fixed to the lungs and the chest wall and diaphragm, respectively. The pleural cavity does not communicate with any other nearby space.

Inspiratory muscles

- **Dominant inspiratory muscle:**
Diaphragm
- **Auxiliary inspiratory muscles:**
 - Neck muscles:**
scaleni, sternocleidomastoid, suprahyoid and infrahyoid
 - Thoracic muscles:**
pectoralis major and minor, external intercostal, subclavius, serratus anterior, levatores costarum
 - Back muscles:**
latissimus dorsi, trapezius (ascending part), levator scapulae, serratus posterior superior and inferior



Expiratory muscles

- **Expiratory muscles involved in forced expiration:**
 - Thoracic muscles:**
internal and innermost intercostals, subcostales, transversus thoracis
 - “Abdominal press” muscles:**
abdominal wall muscles and pelvic-floor muscles

Muscles controlling the vocal ligaments				
Movements of the crico-arytenoid joint		Movements of the cricothyroid joint		
Abduction of the vocal ligaments (widening of the rima glottidis) – respiration position)	Adduction of the vocal ligaments (narrowing of the rima glottidis) – phonation position)	Tightening of the vocal ligaments	Relaxation of the vocal ligaments	Regulation of length and tension of the vocal ligaments
posterior crico-arytenoid	lateral crico-arytenoid	cricothyroid	thyro-arytenoid	vocalis
	thyro-arytenoid			
	transverse and oblique arytenoid			

Muscles controlling the epiglottis	
Ventrocranial movement of the epiglottis (drags the epiglottis to the root of the tongue, opening the laryngeal inlet)	Dorsocaudal movement of the epiglottis (pulls the epiglottis down, closing the laryngeal inlet)
thyro-epiglottic part of thyro-arytenoid	ary-epiglottic part of oblique arytenoid

I. Nose

1. Name the cartilages forming the cartilaginous base of the nose. (p. 206)
2. Name the bony and the cartilaginous structures forming the nasal septum. (p. 207)

II. Nasal cavity

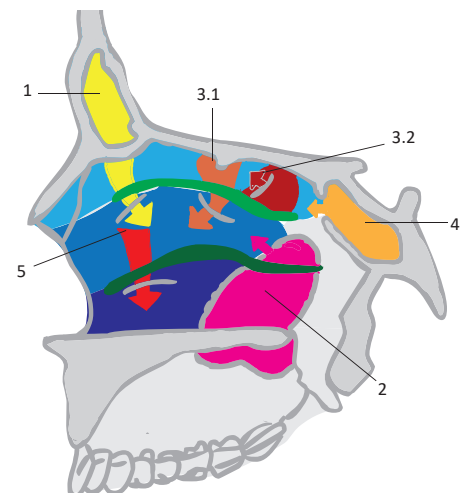
3. List the bones forming the lateral wall of the nasal cavity. (p. 207)
4. Define the term “common nasal meatus”. (p. 208)

III. Pharynx

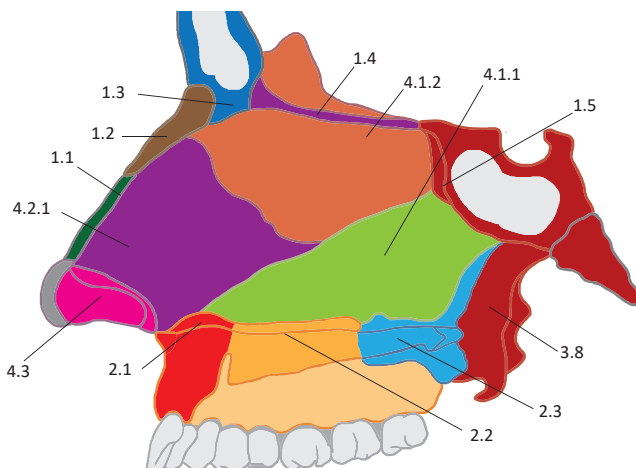
5. State into which part of the pharynx the auditory (Eustachian) tube opens. (p. 209)
6. State into which part of the pharynx the laryngeal inlet opens. (p. 209)

IV. Larynx

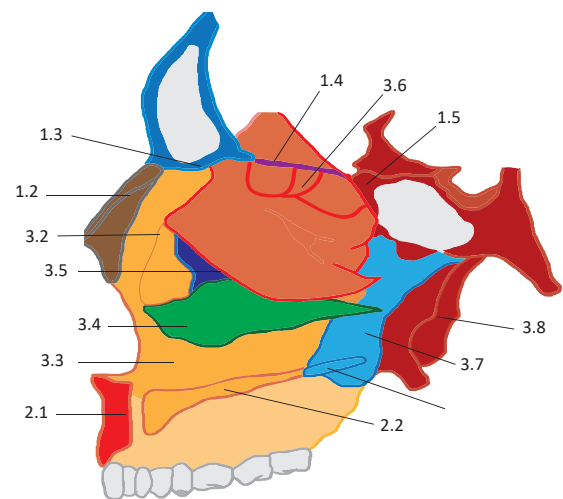
7. List the 5 structures forming the borders of the laryngeal inlet. (p. 210)
8. State the three components of the rima glottidis. (p. 210)
9. State the colour of the vocal fold mucosa during laryngoscopic examination. (p. 210)
10. State the two arteries supplying the larynx. (p. 211)
11. Name the cranial nerve responsible for the viscerosensory, somatomotor and parasympathetic innervation of the larynx. (p. 211)
12. State the direction in which the vocal process of the arytenoid cartilage points. (p. 211)
13. State the direction in which the muscular process of the arytenoid cartilage points. (p. 211)
14. Define the term “quadrangular membrane”. (p. 212)
15. Define the term “conus elasticus”. (p. 212)
16. Explain the function of the posterior crico-arytenoid. (p. 213)



Describe the paranasal sinuses and the nasal meatuses



Describe the structure of the nasal septum



Describe the walls of the nasal cavity

V. Trachea

17. Define the term "carina of trachea". (p. 214)
18. State at which level the tracheal bifurcation is located. (p. 214)
19. Define the term "bronchopericardial membrane". (p. 214)
20. Name the organ that is located dorsally to the trachea. (p. 214)

VI. Lungs

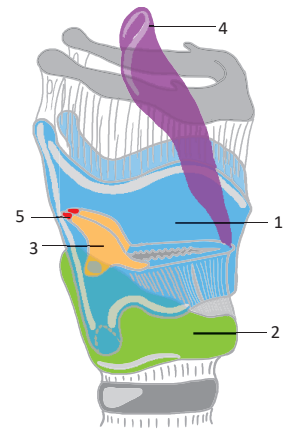
21. Name the part of the airways that ventilates the secondary pulmonary lobule. (p. 217)
22. Describe the topographic relationship between the vagus nerve and the hilum of the lung. (p. 218)
23. Describe the topographic relationship between the phrenic nerve and the hilum of the lung. (p. 218)
24. State the location of the intrapulmonary lymph nodes. (p. 218)
25. State the location of the bronchopulmonary lymph nodes. (p. 218)
26. State the location of the tracheobronchial lymph nodes. (p. 218)
27. Name the 4 parts of the lungs from which lymph is drained into the right lymphatic duct. (p. 219)
28. Name the part of the lung from which lymph is drained into the thoracic duct. (p. 219)

VII. Parietal and visceral pleura

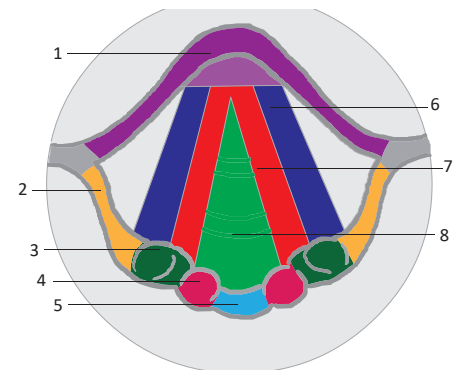
29. List the 4 parts of the parietal pleura. (p. 220)
30. State the deepest part of the pleural cavity. (p. 220)
31. State the approximate physiological amount of pleural fluid in the pleural cavity. (str. 220)

VIII. Mechanism of respiratory muscles

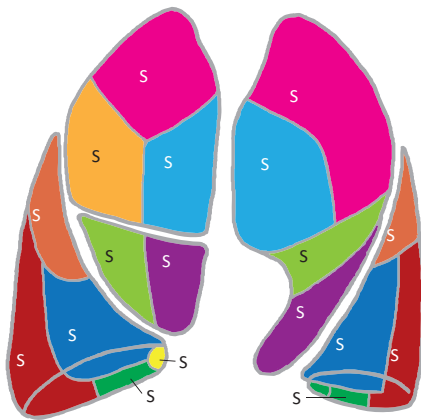
32. Name the main inspiratory muscles. (p. 221)



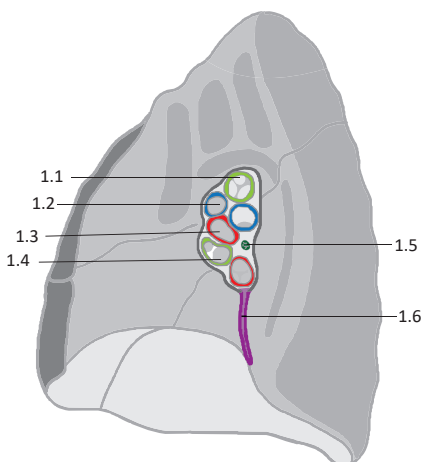
Describe the cartilages of the larynx



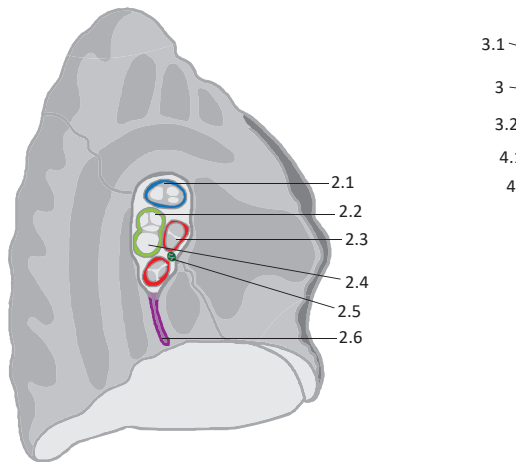
Describe the borders of the laryngeal inlet



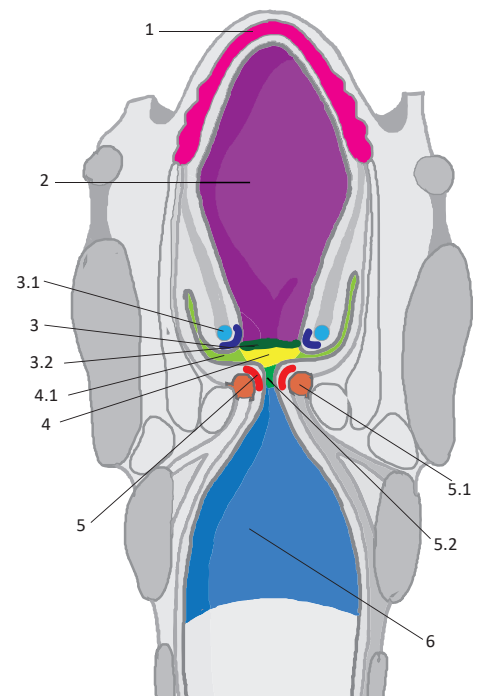
Describe the bronchopulmonary segments



Describe the hilar structures of the right lung



Describe the hilar structures of the left lung



Describe the frontal section of the larynx

We would like to **thank the following physicians, anatomists and medical students** for their invaluable help, devotion and feedback in the preparation of this chapter.

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1. AGUR, A. M. R. and A. F. DALLEY. Grant's Atlas of Anatomy. 13th Ed. Philadelphia: Lippincot Williams & Wilkins, 2012, 888 p. ISBN 978-1-60831-756-1.
2. CHUNG, K. W.; CHUNG, H. M. and N. L. HALLIDAY. BRS Gross Anatomy. 8th Ed. Philadelphia: Lippincott Williams & Wilkins, 2015. 544 p. ISBN 978-1-4511-9307-7.
3. LANZ, T. and W. WACHSMUTH. Praktische Anatomie. 7 bands. Berlin: Springer, 2003, 3658 p. ISBN 978-3-540-40571-9.
4. MILLS, S. E. Histology for pathologists. 4th Ed. Philadelphia: Lippincot Williams & Wilkins, 2012. 1328 p. ISBN-13: 978-1-4511-1303-7.
5. NETTER, F. H. Atlas of Human Anatomy. Professional Edition. 6th Ed. Oxford: Elsevier. 2014. 640 p. ISBN 978-1-455-75888-3.
6. SNELL, R. S. Clinical anatomy by regions. 9th Ed. Philadelphia: Lippincott Williams & Wilkins, 2012. 768 p. ISBN 978-1-60913-446-4.
7. WILLIAMS, P. L., ed. Gray's Anatomy: the anatomical basis of medicine and surgery. 38th Ed. New York: Churchill Livingstone, 1995, 2092 p. ISBN 0-443-05717-6.